

Interpretable dengue detection using XAI-Integrated ensemble and Large Language Models on a reduced-feature clinical dataset

Muhammad Zawad Mahmud, Md. Shihab Reza, Samiha Islam, Shahran Rahman Alve, Riasat Khan *

Electrical and Computer Engineering, North South University, Dhaka 1229, Bangladesh

ARTICLE INFO

Dataset link: [Dengue_ML_Research](#)

Keywords:

Dengue detection
Feature engineering
Machine learning
Ensemble technique
Large Language Model
Permutation feature importance
LIME

ABSTRACT

Dengue fever, a viral disease transmitted by Aedes mosquitoes, is a leading public health threat in populous Southeast Asian countries, especially Bangladesh, Thailand, and India, where transmission of this disease is frequent. This study aims to enhance dengue detection by finding the best classifier for a newly introduced dataset that effectively utilizes a minimal set of features through a comparative analysis of artificial intelligence (AI) techniques integrated with explainable artificial intelligence (XAI) approaches. The private dataset was collected from a local hospital in Dhaka, Bangladesh, containing 1168 samples, among which 639 were male and 529 were female. The dataset comprises 57.1% negative labeled data and 42.9% positive labeled data with various CBC features, including age, gender, hemoglobin, hematocrit, total count (TC), platelet count (PC), and NS1. Several steps have been performed in preprocessing the data for missing values, identifying outliers, and performing multiple feature engineering techniques to maintain the data quality. A range of models spanning classical machine learning (Random Forest, XGBoost, Gradient Boosting, AdaBoost, LightGBM, etc.), deep learning (1D-CNN, MLP, and ANN), and recent tabular foundation models (TabPFN, TabICL, FT-Transformer, SAINT, TabNet, TabM, and TabR) were developed. Ensemble techniques, stacking, bagging, and blending classifiers are adopted, while hyperparameters are tuned using GridSearchCV and Keras Tuner. Four large language models (LLMs) BERT, BioBERT, ClinicalBERT, and distilBERT, were fine-tuned after transforming the tabular records into text, and two ensemble LLMs were constructed. XAI techniques were incorporated to add interpretability. LIME, SHAP, and Permutation Feature Importance (PFI) are applied to explain feature contributions to the model across the entire dataset or test set. The stacking classifier achieved quite strong performance with 89.74% accuracy and an F1 score of 0.880 with low computation cost. The fine-tuned BERT achieved the best overall results, with 95.73% accuracy and an F1 score of 0.940. These findings illustrate the promise of LLM-based approaches for accurate and interpretable dengue detection from reduced-feature tabular data.

1. Introduction

Dengue fever is a widespread and viral disease transmitted by mosquitoes that poses a global health threat, especially in tropical and subtropical regions [1]. The dengue virus, responsible for this disease, is transmitted mainly by four species of Aedes aegypti mosquito [2]. It breeds in containers with stagnant water, and this kind of mosquito flourishes in urban and semi-urban areas. Infection with the virus can be mild, resulting in flu-like symptoms, or severe, including dengue hemorrhagic fever and dengue shock syndrome, which may be fatal without proper treatment. As of early 2024, there have been over 13 million globally reported cases of dengue and over 8500 related deaths, with the most significant numbers seen in the WHO PAHO region, where greater than 11 million overall cases were reported (53%

laboratory-confirmed). Eight of the 13 countries that reported cases between 2016 and 2020 were present in this group. Brazil was the most affected (9.5 M), followed by Argentina, Mexico, Paraguay, and Colombia [3].

The consequences of dengue are especially devastating in Bangladesh, where high population density coupled with humidity and temperature favors Aedes breeding [4]. When dengue has been endemic in Bangladesh, it has historically occurred near the three largest cities, Dhaka, Chittagong, and Khulna, where high urban density and favorable environmental conditions drive transmission [5]. In Bangladesh, where healthcare data is frequently inconsistent or inadequately maintained, recent data has highlighted dengue as a severe threat to public

* Corresponding author.

E-mail address: riasat.khan@northsouth.edu (R. Khan).

<https://doi.org/10.1016/j.array.2026.101120>

Received 8 August 2025; Received in revised form 30 July 2026; Accepted 30 July 2026

Available online 5 August 2026

2590-0056/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

health. Dengue has spread on an epidemic scale, with more than 203,406 cases and 989 fatalities between January and September 2023, amounting to a case fatality rate of 0.49% [6]. Between 1 January and 30 October 2024, a total of 72,393 dengue cases and 355 deaths were reported, indicating an urgency for better surveillance and control. 95% or more of infections and deaths occurred between July and September, corresponding to several factors, e.g., population density, rainfall, air quality, etc.

Machine learning (ML) provides an innovative and integrative method for the efficient detection and classification of dengue that serves to transform efficient approaches involved in disease management and control. Using algorithms to assess large datasets, including patient medical records, weather patterns, and mosquito population levels, ML models can even predict outbreaks with notable accuracy. For instance, Jannah et al. [7] reported that the fuzzy K-nearest neighbor model achieved an 85.01% accuracy in dengue prediction in a study conducted with a sample size of 300 instances at UCI. Similarly, Nordin et al. [8] found that the Support Vector Machine (SVM) model reached an 85% accuracy using data from 84 cases at the Health Department in Malaysia. Moreover, Davi et al. [9] demonstrated that an artificial neural network (ANN) could predict dengue with 86% accuracy using data from 102 cases at three local hospitals in Recife.

In this work, we have built a private dataset by collecting information from a local hospital in Dhaka, Bangladesh. The dataset contains different features of the complete blood count (CBC) data for 1168 individuals, classified as dengue ‘positive’ or ‘negative,’ and six attributes. The dataset has been preprocessed employing diverse techniques, and feature engineering techniques are applied to create three more features from existing ones. Various machine learning models have been applied, i.e., Random Forest, LightGBM, XGBoost, AdaBoost, Gradient Boosting, etc., with a stacking classifier. We deployed three deep-learning models, a 1D-CNN, an ANN, and an MLP with a suite of recent tabular foundation and deep-learning models (TabPFN, TabICL, FT-Transformer, SAINT, TabNet, TabM, and TabR). Popular LLMs, including BERT, BioBERT, ClinicalBERT, and distilBERT, were fine-tuned and combined into two ensemble LLM classifiers. All the model’s hyperparameters are tuned using GridSearchCV and Keras Tuner. The role of individual features in the decision-making of these black-box models is investigated using explainable AI approaches, namely LIME, SHAP, and PFI. The model predicts NS1 antigen positivity directly from complete blood count (CBC) features. Its intended use is early triage support rather than diagnosis or replacement of laboratory testing. A CBC-based estimate of NS1 positivity can help clinicians prioritize suspected dengue patients for confirmatory NS1 testing and closer monitoring while awaiting laboratory results. The study offers several significant contributions, summarized as follows:

- Introducing a private dataset of 1168 possible dengue patients with complete blood count (CBC) features — age, gender, hemoglobin, hematocrit, total count (TC), and platelet count (PC) — labeled with NS1 status.
- Deploying feature engineering to derive three additional features, hemoglobin_level, hematocrit_level, and total_pc_interaction, from hemoglobin, hematocrit, total count, and platelet count.
- Construction of a stacking ensemble model and a blending ensemble model from AdaBoost as a meta-classifier, along with XGBoost, Random Forest, and Gradient Boosting as multilayer perceptron-based learners for blending and XGBoost, Random Forest, and bagging for stacking and benchmarking it against recent tabular foundation and deep-learning models (TabPFN, TabICL, FT-Transformer, SAINT, TabNet, TabM, TabR).
- Fine-tuning recent LLMs (BERT, BioBERT, ClinicalBERT, distilBERT) after transforming tabular records into text with appropriate data augmentation.
- Building two ensemble LLM classifier that takes the predicted labels of the models, uses best among majority voting, weighted-average voting, median voting, and product-rule voting and produces outputs.

- Applying explainable AI techniques (LIME, SHAP, and PFI) to interpret model behavior and the contribution of each feature.

The novelty of this work lies in introducing a private dataset and integrating explainable ensemble, LLM, and transformer models with an understanding of the roles of the features in the decision-making of dengue detection. Previously, very few studies integrated LLMs to do classification studies from tabular datasets.

The rest of the article is as follows: Section 2 and Section 3 describe the related works and methodology, respectively, whereas Section 4 analyzes the results, and Section 5 discusses the conclusion and future directions.

2. Related work

Dengue fever is spreading worldwide as a pandemic phenomenon. Many researchers have attempted to develop automatic diagnostic models by introducing machine learning algorithms to reduce global incidence. A few recent articles on automatic dengue detection have been briefly discussed in the subsequent paragraphs. For instance, Tuan and Dang [10] applied ML models such as Random Forest, Gradient Boosting, and LSTM to measure the effect of climate factors on dengue, acute diarrhea, HFMD, and influenza in Ba Ria-Vung Tau, Vietnam. Lu et al. [11] developed a model combining MLR, LSTM, and SI-SIR to forecast dengue cases in Selangor, Malaysia, using climate variables.

Gupta and Bhattachakosol [12] used Bayesian regularization neural networks to solve a fractional-order dengue viral model, showing that fractional derivatives provide more accurate results than integer ones. Dey et al. [13] predicted the number of dengue incidences in Bangladesh on a dataset called “DengueBD.” The authors utilized an extensive ecological and social determinant network to enrich the model prediction analyses involving 11 districts. In a comparison of two primary regression models, multiple linear regression (MLR) and support vector regression (SVR), the SVR model was the best performer with a prediction accuracy of 75% and MAE of 4.95 compared to an MLR model, which showed an accuracy level of 67%.

In a recent work by Riya et al. [14], the authors propose an innovative technique for early detection and prediction of dengue fever using complete blood count (CBC) data from patients in Bangladesh. The research was based on an enormous outbreak during 2023, and it analyzed the dataset, including 322 samples, using multiple machine learning models coupled with diverse feature selection methods. After doing hyperparameter tuning using GridSearchCV, the stacking classifier was able to achieve an accuracy of 96.88% and a macro F1 score of 0.9646, with LightGBM as the meta-classifier. Sarma et al. [15] utilized clinical data from two large medical centers in Bangladesh to develop an ML model for predicting dengue fever outcomes. The authors used a dataset of 209 patients (divided into normal fever, normal dengue, and severe dengue) with a training-to-test ratio of 70:30. The authors assessed patient diagnosis reports, medical history, and symptoms using decision trees and random forest algorithms. The decision tree algorithm outperformed with an optimal average accuracy of 79%.

Akter et al. [16] demonstrated the predictive power of machine learning over traditional time series methods. The authors employed data on dengue patients between January 2016 and July 2021 collected from the WHO and DGHS. The applied custom CNN model with a five-hidden-layer (neurons) neural network accomplished the lowest error of 0.003. The ARIMA model forecasts the dengue outbreaks by analyzing the time series using 15.29 MAPE. The average MAPE was lower for all other models, but a neural network model obtained the best performance with the lowest MAPE of 1.15. Jannah et al. [7] analyzed the categorization of Dengue Hemorrhagic Fever (DHF) with Fuzzy K-Nearest Neighbor (FKNN), which appeared to increase diagnostic accuracy. The analysis showed optimizing ‘k’ selection across multiple tries and identifying accuracies at k values between 3–20.

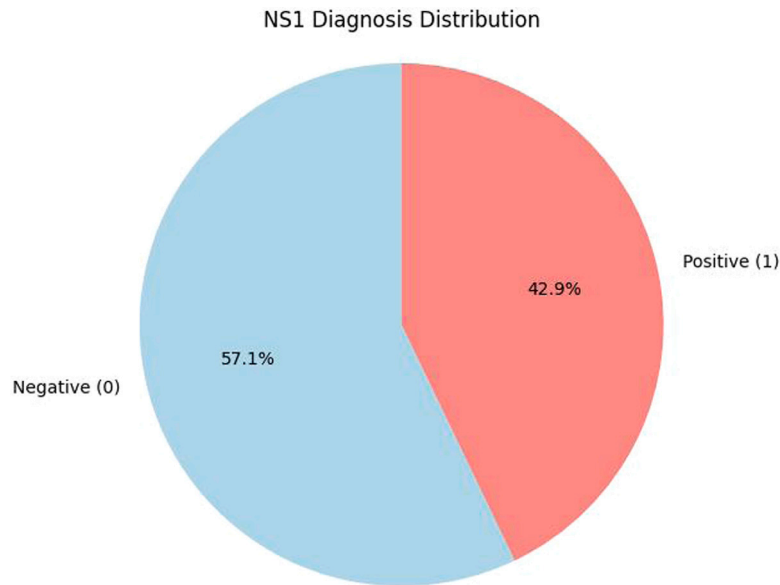


Fig. 1. Number of samples for each label in the collected dataset.

They utilized a balanced dataset with two classes having 300 cases, and reachability analysis is permitted to represent all conditions in patients' data qualitatively. Optimally, the FKNN method performed at $k=13$ with an accuracy of 85.01% (maximum), indicating an improved ability to classify DHF versus other options within this range.

Majeed et al. [17] applied a series of LSTM-based models with spatial and temporal attention mechanisms to forecast dengue fever cases in Malaysia. The performance of their proposed stacked LSTM approach with a spatial attention model provided the lowest average RMSE value, 3.17, and achieved better results than other benchmarks, i.e., SVM, DT, and ANN. Nordin et al. [8] established a powerful forecasting potential by combining SVM models based on different kernel functions. They concluded that the classification performance of their study was applied to datasets obtained from the Kelantan Health Department after applying data scaling, which is to convert data into a specific range, i.e., (0, 1), before feeding it to model training. The best model was the SVM with an RBF kernel, giving a prediction accuracy of 85%.

Salim et al. [18] demonstrated the promising capability of machine learning techniques in public health surveillance by exemplifying their effectiveness in forecasting dengue outbreaks within Selangor, Malaysia. The authors modeled the predictability of five top-differential-incidence districts during 2013–17 using climate variables, including temperature, specific humidity, etc. The SVM model depicted better performance with 70% accuracy and high specificity of 95% but low precision of 56%. Chaw et al. [19] synthesized more details on the application of various machine learning models predicting the risk of shock development in dengue patients using an electronic records dataset. The applied bagging algorithm outperformed traditional machine learning algorithms such as decision trees, logistic regression, SVM, and neural networks, with a 14.5% performance improvement over the worst individual decision tree model in this experimental study. The decision tree model was the best-performing one among all the applied machine learning models and achieved an F1 score of 0.92 with an AUC of 0.64.

Mello-Román et al. [20] applied ANN with SVM techniques for dengue diagnosis using high-incidence data from Paraguay. The authors showed phenomenal performance improvements compared to the baseline deep CNNs in their optimized network setup. The ANN multilayer perceptron model attained an average accuracy of 96% with sensitivity and specificity rates of 96% and 97%, respectively. Davi et al. [9]

introduced a novel hybrid strategy of SVM and ANN models for automatic dengue fever classification. The authors used human genome data and identified 322 SNPs (single nucleotide polymorphisms) related to innate immunity. Surprisingly, the ANN model derived to predict levels of dengue fever severity with 13 top immune SNPs achieved a median performance exceeding an accuracy level of 86%, with sensitivity and specificity rates at 98% and 91%, respectively.

Fernández et al. [21] established a clinical, hematology, and demographic-based model by including 548 patients from Honduras to distinguish dengue from other febrile conditions. Retro-ocular pain, petechiae, and gingival bleeding were positive predictors of dengue, while epistaxis and pale skin had negative associations with the disease. High sensitivity values have already been found, but the implemented logistic regression model attained a specificity of 77.2% and an accuracy of 69.2%, highlighting the need for further validation and enhancement of predictive models. Prome et al. [22] forecasted dengue cases in Bangladesh employing diverse ML models, including ensemble learning methods. Support Vector Regression with corresponding optimized hyperparameters achieved better accuracy prediction results with the lowest mean absolute error value of 4.112 and root mean squared error value of 16.04.

Although the reviewed literature employed various machine learning techniques with excellent performances to predict dengue with significant results, several critical limitations exist. Most current research is based on public datasets that might not fully represent the data complexity. There is a general lack of advanced feature engineering and explainability, which restricts both the interpretability and broader applicability of the applied models. To overcome these gaps, we address these limitations by introducing a novel private dataset with extensive clinical and hematological attributes and corresponding feature engineering strategies. Furthermore, we develop robust ensemble models using stacking techniques evaluated under a nested cross-validation protocol with hyperparameters selected by inner cross-validation, large language models, and enhanced with state-of-the-art explainable AI methods. The integrated approach enhances predictive accuracy and offers a clearer understanding of how the model arrives at its decision.

3. Methodology

This section represents the methodology part of the proposed dengue classification system.

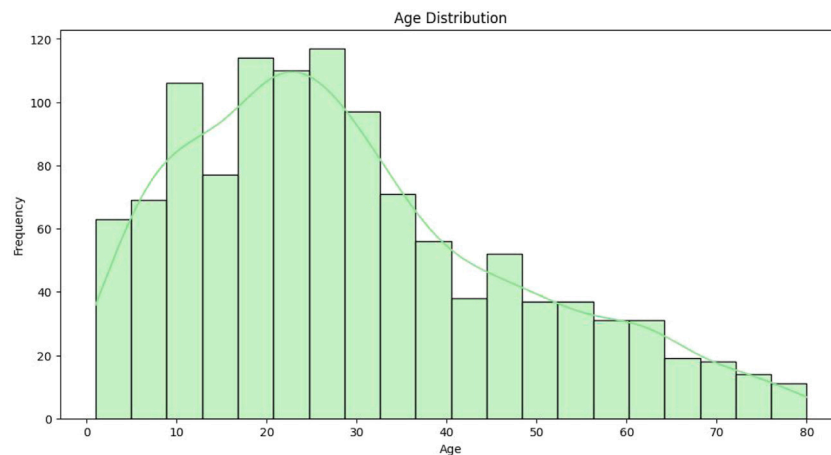


Fig. 2. Age distribution of the curated dataset.

3.1. Dataset

This subsection describes how the private dengue dataset was formed and preprocessed. The dataset was explored.

3.1.1. Collection of the dataset

The dataset used in this study is a private collection obtained from a local hospital in Dhaka, Bangladesh. Data were collected during the monsoon season (July–August 2024), a period typically associated with heavy rainfall and a peak in dengue cases in the region. All data were collected anonymously in compliance with ethical standards and under the guidelines approved by the Institutional Review Board (IRB) of North South University, Dhaka, Bangladesh. All participants provided written informed consent prior to data collection. The dataset comprises 1168 samples, including 639 male and 529 female patients. The age range of the subjects spans from 20 to 80 years. Each sample includes six features: age, gender, hemoglobin, hematocrit, total counts (TC), and platelet counts (PC), along with a binary label (NS1) indicating either a positive or negative dengue diagnosis. Fig. 1 presents the distribution of positive and negative labels.

3.1.2. Dataset preprocessing

During preprocessing of the collected data, we first converted the categorical features (such as ‘Gender’ and ‘NS1’) into numerical form. The values for ‘Gender’ were converted (mapped) internally to 0 for female and 1 for male, whereas the same mapping was done to NS1, put into (0 for negative and 1 for positive results) so that the machine learning algorithms can be applied. The median of each column is used to replace missing values across the dataset. All the numerical columns were validated and transformed to be numeric-ready. This extensive preprocessing helps reduce potential biases and errors found during model training and evaluation.

3.1.3. Exploring the dataset

Fig. 2 shows an age distribution histogram combined with a line graph with ages on the x-axis and frequency on the y-axis. The histogram shows that age groups ranging from 20 to 40 are most prevalent, indicating a young population. This line graph is the distribution curve, meaning the data points fit a theoretical distribution, peaking around 30 years and tailing off as age increases.

Fig. 3 illustrates the gender distribution within the collected dengue dataset. Out of 1168 samples, 639 were male, and 529 were female, indicating a slightly higher representation of male patients.

Fig. 4 presents the age-wise distribution of NS1 diagnostic outcomes. The majority of positive NS1 cases were observed among individuals aged between 10 and 30 years. While both positive and negative

cases spanned the full age range of 0 to 80, younger age groups showed a higher frequency of positive diagnoses.

Fig. 5 shows the distribution of NS1 diagnostic outcomes across genders. Among both male and female participants, the number of NS1-negative cases is higher than the number of NS1-positive cases. Male participants exhibited a greater number of both positive and negative cases compared to female participants.

The correlation heatmap shown in Fig. 6 illustrates the relationships between some of the major physiological markers and the test result for NS1 dengue virus antigen within our dataset. Hemoglobin and hematocrit were insignificantly correlated with each other, as well as with the NS1 diagnosis for hemoglobin. Neither of these variables shares a linear relationship with the presence of the NS1 antigen, yet the variables are capable of numerical behavior interchangeably. For example, the platelet counts (PC) have a low-to-moderate negative correlation with the NS1 test result (-0.35), implying that a positive NS1 test may be associated with lower PC, an important clinical feature of dengue infection. The total count (TC) of leukocytes has a very low correlation with NS1, suggesting that in the employed dataset, TC is less likely to indicate the infection status than platelet count.

3.2. Feature engineering

In this work, a new feature was created to categorize hemoglobin and hematocrit levels as low, normal, or high based on established medical guidelines that account for gender-specific thresholds. According to UCSF Health, normal hemoglobin levels range from 13.8 to 17.2 grams per deciliter (g/dL) for males and 12.1 to 15.1 g/dL for females [23]. Similarly, the NCBI’s Clinical Methods reports normal ranges of 14–18 g/dL for males and 12–16 g/dL for females [24]. Standard medical guidelines define normal hematocrit ranges as 41%–50% for males and 36%–44% for females [25]. Values falling below or above these thresholds are classified as low or high, respectively. A total count-to-platelet count interaction feature was derived by multiplying total blood count (TC) and platelet count (PC), aiming to capture complex interactions that may be critical for improving dengue diagnosis. Aligning classification thresholds with these widely accepted medical standards enhances the clinical relevance, precision, and reliability of the analysis.

3.3. Data conversion for LLM

The conversion of structured dengue data parameters into natural-language sequences made it possible to apply transformer-based modeling for hematological profiles. The Python pandas library was used to transform each patient record containing numerical features (age,

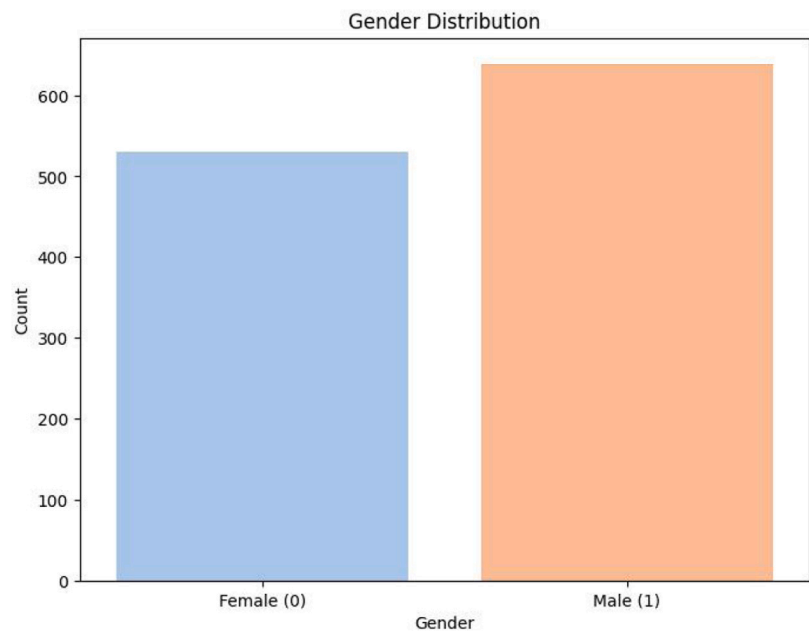


Fig. 3. Gender distribution of the employed dataset.

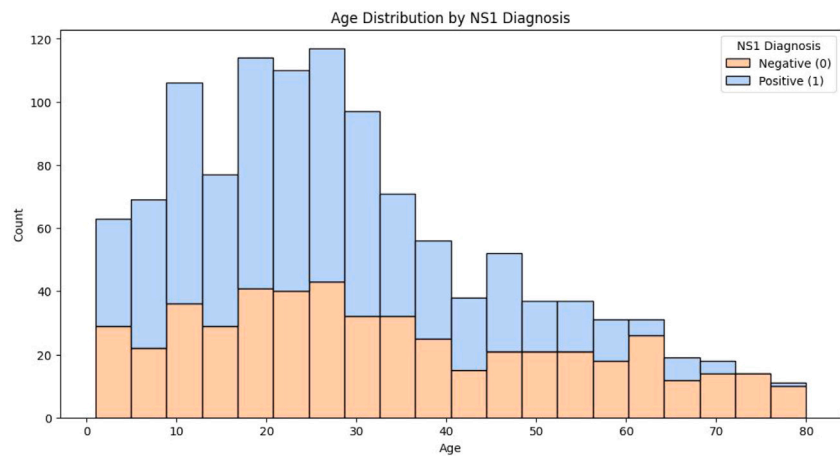


Fig. 4. Age distribution with NS1 label of the employed dataset.

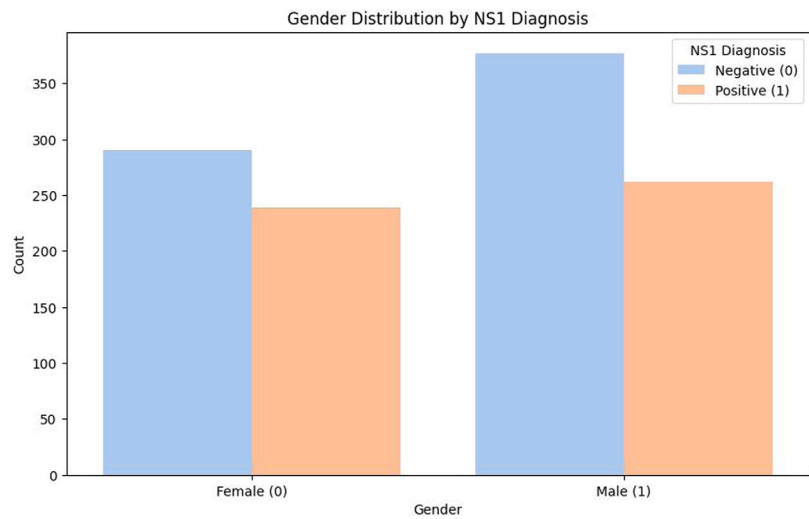


Fig. 5. Gender distribution with NS1 label of our dataset.

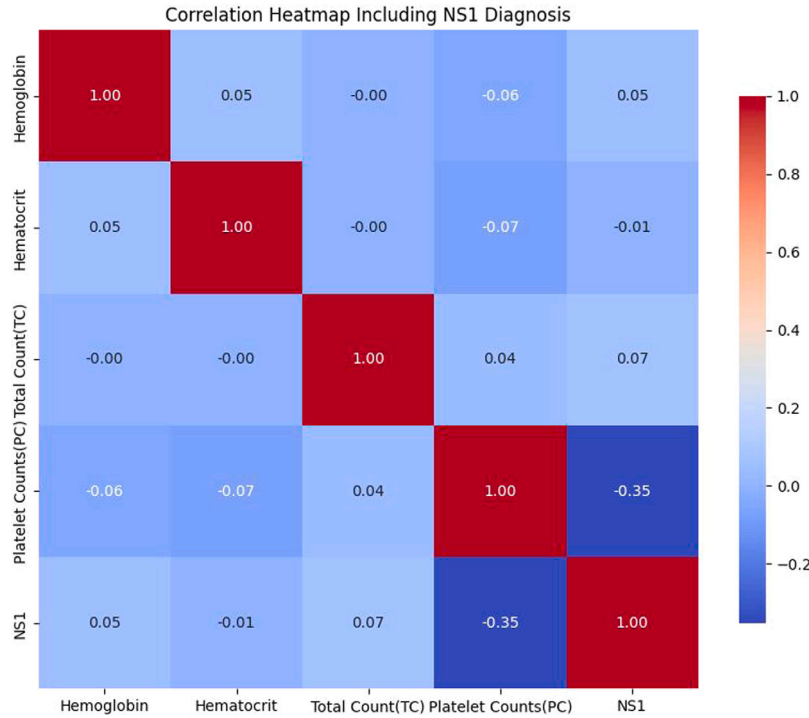


Fig. 6. Heat map of various numerical features of the collected dengue dataset.

hemoglobin, hematocrit, total cell count, and platelet count) and derived biomarkers (hemoglobin level, hematocrit level, and total-PC interaction) into text. The data integrity was maintained by explicitly marking missing values as “Not available.” The encoding of each record followed this template:

A age-year-old individual with a hemoglobin level of value g/dL (category), hematocrit of value % (category), total cell count of value, platelet count of value, total-PC interaction: value, tested result for NS1.

The NS1 label was held out separately as the supervised classification target. It was provided only as the ground-truth label during fine-tuning and only for scoring at test time. A representative serialized instance used for both training and inference is the following:

A 39.0-year-old individual with a hemoglobin level of 8.2 g/dL (1), hematocrit of 31.1% (1), total cell count of 9.03, platelet count of 102, total-PC interaction: 921.06.

3.4. Data augmentation

To enhance sample diversity and mitigate model overfitting and underfitting, we expanded the dengue text corpus fourfold using the NLP augmentation (NLPAug) library. The train set’s original ($N = 934$) sentences received three different augmentation strategies, which produced three new variants through WordNet synonym replacement, random token swapping, and bert-base-based masked language model contextual word substitution. We made sure no clinical text or number was altered by using mask. The system handled exceptions by replacing failed augmentations with the original sentence to preserve the corpus structure. The total dataset reached $4 \times N$ samples after this process. The augmented corpus received training and fine-tuning from transformer architectures BERT, BioBERT, ClinicalBERT, and distilBERT to improve their performance in dengue classification.

3.5. Dataset splitting

After completing the preprocessing and feature engineering, the dataset was split by a 90–10 ratio, where 90% of the data was used for training and 10% for testing. This approach was followed for machine learning and deep learning models. While training, 5-fold cross-validation and GridSearchCV were applied to ensure that the model’s performance was evaluated robustly and to optimize the hyperparameters to improve its performance. The best classical machine learning models were evaluated by nested cross-validation on the full dataset. The LLMs, which require a held-out validation set for checkpoint selection, instead used a fixed 80–10–10 split: 80% of the data for training, 10% for validation, and 10% for testing.

3.6. Machine learning model training

This subsection describes how the models were trained in this work.

3.6.1. Random forest

For training the Random Forest [26] classifier, we set up a parameter grid and used GridSearchCV to search for hyperparameters based on cross-validation. When the parameters were found, we initialized a new Random Forest model with these parameters and then fit it to the training data, which were scaled using “StandardScaler”. A distinct test dataset is utilized to assess the model’s performance in terms of accuracy and generalizability, confirming strong predictive abilities.

3.6.2. XGBoost

We trained the XGBoost [27] ensemble model with a 5-fold cross-validation-integrated GridSearchCV hyperparameter tuning framework. A new instance of an XGBoost model was instantiated and fitted to the scaled training data after determining the optimal parameters. The performance of the model was evaluated using the accuracy on a separate test dataset and then validated through 5-fold cross-validation.

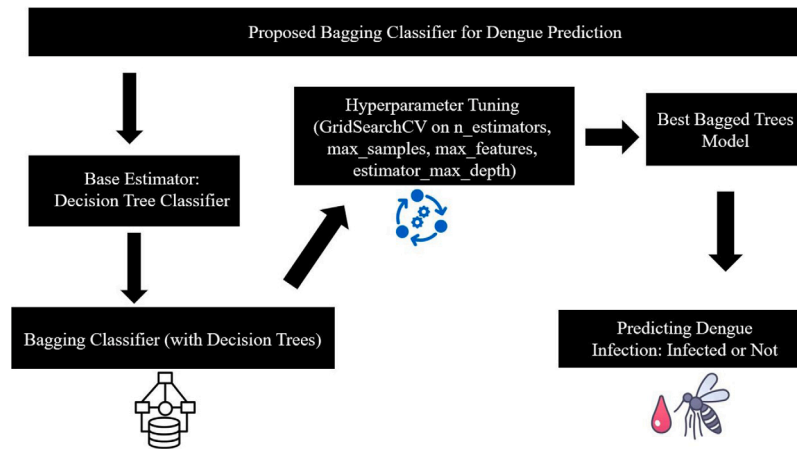


Fig. 7. Model Architecture of the Proposed Bagging Classifier.

3.6.3. Gradient boosting

The gradient boosting [28] was trained by tuning some of the critical hyperparameters (max depth, estimators, learning rate, and subsample) using a 5-fold cross-validation with GridSearchCV. A scaled training dataset was then used to fit the best model and make predictions on training and test sets.

3.6.4. AdaBoost

We trained the AdaBoost [29] classifier using GridSearchCV, which performed hyperparameter optimization on ROC-AUC scoring. Once we had the perfect parameters, we trained the AdaBoost model using scaled training data. We computed the accuracy of the training dataset and testing dataset, then validated it by 5-fold cross-validation.

3.6.5. LightGBM

To achieve good predictive performance and have a robust model across subsets of data, LightGBM [30] was trained by fine-tuning the hyperparameters of this model via defining a grid of parameters and GridSearchCV using 5-fold cross-validation to find and fit the best combination. Once the best parameters were found, the model was trained using those optimal parameters on the training dataset for fast and efficient learning. In the end, we validated the model (cross-validation and test accuracy) and checked for the suitability of the task using different metrics (confusion matrix and classification report).

3.6.6. Bagging classifier

In the Bagging Classifier, the Decision Tree model is used as a base estimator. Hyperparameter tuning was conducted via GridSearchCV with 5-fold cross-validation across various hyperparameters, e.g., number of estimators, maximum tree depth, and fractions of samples and features used for each tree. The best configuration found was 'n_estimators': 50, 'max_samples': 0.5, 'max_features': 0.8, and 'max_depth': None. The corresponding out-of-fold predictions are passed through a k-fold blending process and fed into an AdaBoost meta-classifier to determine whether a patient is infected with dengue or not. The detailed pseudocode of the applied Bagging Classifier is displayed in Algorithm 1. The proposed model architecture is represented in Fig. 7.

3.6.7. Stacking classifier

In this work, we have developed a stacking classifier with a Bagging Classifier, XGBoost, and Random Forest as base models and AdaBoost as the meta-model. For each model, the hyperparameters were tuned with GridSearchCV. The stacking classifier used stratified K-fold cross-validation on the training set and made predictions on the test stage.

Algorithm 1 Proposed Bagging Classifier for Dengue Prediction

- 1: **Base Estimator and Bagging Setup:**
- 2: Initialize a `DecisionTreeClassifier` with `random_state=42` as the base estimator
- 3: Instantiate a `BaggingClassifier` using the base estimator and set `random_state=42`
- 4: **Hyperparameter Tuning:**
- 5: Define the hyperparameter grid for `BaggingClassifier`:
- 6: `n_estimators: {10, 50, 100}`
- 7: `max_samples: {0.5, 0.8, 1.0}`
- 8: `max_features: {0.5, 0.8, 1.0}`
- 9: `estimator__max_depth: {5, 10, None}`
- 10: Perform 5-fold GridSearchCV using accuracy as the scoring metric
- 11: Select the best hyperparameters based on cross-validation performance
- 12: **Training and Evaluation:**
- 13: Train the tuned `BaggingClassifier` on the training set ($X_{\text{train_scaled}}, Y_{\text{train}}$)
- 14: Predict on the test set ($X_{\text{test_scaled}}$)
- 15: Compute evaluation metrics: Accuracy, Precision, Recall, and F1 score
- 16: Generate classification report and construct a confusion matrix (visualized with a heatmap)
- 17: **Result:**
- 18: Return the tuned `BaggingClassifier` along with evaluation metrics, cross-validation scores, and confusion matrix visualization

The detailed pseudocode of the proposed stacking classifier is displayed in Algorithm 2. The proposed model architecture is represented in Fig. 8.

3.6.8. Blending classifier

A blending Classifier was built, comprising Gradient Boosting as a base model, and then we did the same for XGBoost and Random Forest, with simultaneous creation for the other models as Blenders above. The ground truth features were combined by making meta-features via K-Fold cross-validation on the train set, and these models were fine-tuned using GridSearchCV. The relevant meta-features were passed to the AdaBoost meta-model, which serves as the final predictor. An ensemble (voting classifier) was trained and evaluated to combine the predictions of diverse models with predictive power. The detailed

Algorithm 2 Proposed Stacking Classifier for Dengue Prediction

Require: Preprocessed training set ($X_{\text{train_scaled}}, Y_{\text{train}}$), test set ($X_{\text{test_scaled}}, Y_{\text{test}}$), number of folds K (here, $K = 5$)

Ensure: Trained stacking classifier and corresponding evaluation metrics

- 1: **Hyperparameter Tuning:**
- 2: All models are initialized with `random_state=42` for reproducibility.
- 3: **Bagged Trees:** Tune a `BaggingClassifier` (base: `DecisionTreeClassifier`) using a grid of parameters:
- 4: [`n_estimators`, `max_samples`, `max_features`, `estimator__max_depth`] via 5-fold `GridSearchCV` (scoring = accuracy) $\rightarrow$ `best_bagged_trees`
- 5: **XGBoost:** Tune `XGBClassifier` with parameters:
- 6: [`max_depth`, `n_estimators`, `learning_rate`, `colsample_bytree`, `subsample`] via 5-fold `GridSearchCV` (scoring = ROC AUC) $\rightarrow$ `best_xgb`
- 7: **AdaBoost:** Tune `AdaBoostClassifier` using parameters:
- 8: [`n_estimators`, `learning_rate`, `algorithm`] via 5-fold `GridSearchCV` (scoring = ROC AUC) $\rightarrow$ `best_ada`
- 9: **Random Forest:** Tune `RandomForestClassifier` using parameters:
- 10: [`n_estimators`, `max_features`, `max_depth`, `criterion`] via 5-fold `GridSearchCV` (scoring = ROC AUC) $\rightarrow$ `best_rf`
- 11: **Stacking Classifier Construction:**
- 12: Define base learners: `best_rf`, `best_xgb`, `best_bagged_trees`
- 13: Define meta-learner: `best_ada`
- 14: Instantiate `StackingClassifier` with the defined base learners and meta-learner, using:
- 15: `StratifiedKfold` ($K = 5$), `stack_method='predict_proba'`, and `n_jobs=-1`
- 16: **Model Training and Evaluation:**
- 17: Train the stacking classifier on ($X_{\text{train_scaled}}, Y_{\text{train}}$)
- 18: Predict on $X_{\text{test_scaled}}$
- 19: Compute evaluation metrics
- 20: Generate a classification report and a confusion matrix
- 21: **Return:** Trained stacking model and all evaluation metrics

pseudocode of this method is displayed in Algorithm 3. The proposed model architecture is represented in Fig. 9.

3.7. Deep learning model training

Deep Learning models have shown promising performance for tabular data in the past. Here, we state the training process of the applied deep learning models.

3.7.1. MLP

We initiated the training of the Multilayer Perceptron (MLP) classifier [31] by choosing an MLP with a fixed structure and hyperparameters, and further, optimized a range of hyperparameters that include layer sizes, activation functions, solvers, regularization rates, and learning strategies via `GridSearchCV` and using 5-fold cross-validation.

3.7.2. 1D-CNN

After scaling the features to uniformity using `StandardScaler`, the 1D Convolutional Neural Network (CNN) model [32] was trained for classifying dengue data. The input features were reshaped to a three-dimensional format due to the 1D CNN input requirement. The proposed 1D-CNN model architecture had two convolutional layers with ReLU activation, followed by max pooling and flattening, a fully connected dense layer (with an additional dropout regularization layer

Algorithm 3 Proposed Blending Classifier for Dengue Prediction

- 1: **Base Model Tuning:**
- 2: **Gradient Boosting:** Tune `GradientBoostingClassifier` using grid {`max_depth`, `n_estimators`, `learning_rate`, `subsample`} via 5-fold `GridSearchCV` (scoring = ROC AUC) $\rightarrow$ `best_gb`
- 3: **XGBoost:** Tune `XGBClassifier` using grid {`max_depth`, `n_estimators`, `eta`, `colsample_bytree`, `subsample`} via 5-fold `GridSearchCV` (ROC AUC) $\rightarrow$ `best_xgb`
- 4: **Random Forest:** Tune `RandomForestClassifier` using grid {`n_estimators`, `max_features`, `max_depth`, `criterion`} via 5-fold `GridSearchCV` $\rightarrow$ `best_rf`
- 5: **Blending Feature Construction:**
- 6: Initialize `blending_X_train` [`n_train`, 3] and `blending_X_test` [`n_test`, 3]
- 7: Perform 5-fold cross-validation:
- 8: For each fold:
- 9: Train `best_gb`, `best_xgb`, `best_rf` on the training fold
- 10: Generate out-of-fold probability predictions for `blending_X_train`
- 11: Compute and average test set predictions across folds for `blending_X_test`
- 12: **Meta-model Tuning and Evaluation:**
- 13: Tune `AdaBoostClassifier` (meta-model) using grid {`n_estimators`, `learning_rate`, `algorithm`} via 5-fold `GridSearchCV` (ROC AUC) $\rightarrow$ `best_ada`
- 14: Predict on `blending_X_test` using `best_ada`
- 15: Compute evaluation metrics
- 16: Generate classification report and visualize confusion matrix using a heatmap
- 17: **Return:**
- 18: Trained blending meta-model (`best_ada`) and all evaluation results

to reduce overfitting), and a sigmoid-activated output node for binary classification. The model was trained for 20 epochs with a batch size of 32 using the Adam optimizer and binary cross-entropy loss. The model was validated against the test set to prevent any performance drop. The total training time for the model was 4.32 s, with a training time per epoch of 0.22 s. The total model size was 245.98 KB, and the total parameters were 17217.

3.7.3. ANN

The Artificial Neural Network (ANN) [33] was trained to classify dengue. The features were scaled using `StandardScaler` before being divided into train and test sets and fed into the ANN model. The model's architecture consisted of three hidden layers with ReLU activation functions, dropout layers to prevent overfitting, and a sigmoid activation to get probabilities for binary classification at the end. The training was done with the Adam optimizer, a binary cross-entropy loss function, and an accuracy metric over 20 epochs with a batch size of 32 while validating against the test set to prevent any performance drop. The total training time for the model was 3.81 s, with a training time per epoch of 0.19 s. The total model size was 290.27 KB, and the total parameters were 20929.

3.8. Tabular foundation model benchmark

Recent architectures designed specifically for tabular data have challenged the long-standing dominance of gradient-boosted trees. TabPFN, a prior-data-fitted transformer, performs classification in a single in-context forward pass and has matched or exceeded boosted trees on small tabular datasets without per-dataset training [34]. To place our results in this context, we benchmarked seven such models

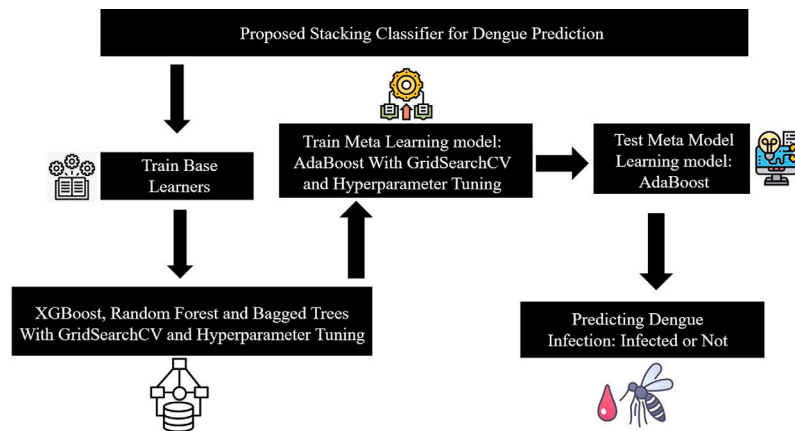


Fig. 8. Model Architecture of the Proposed Stacking Classifier.

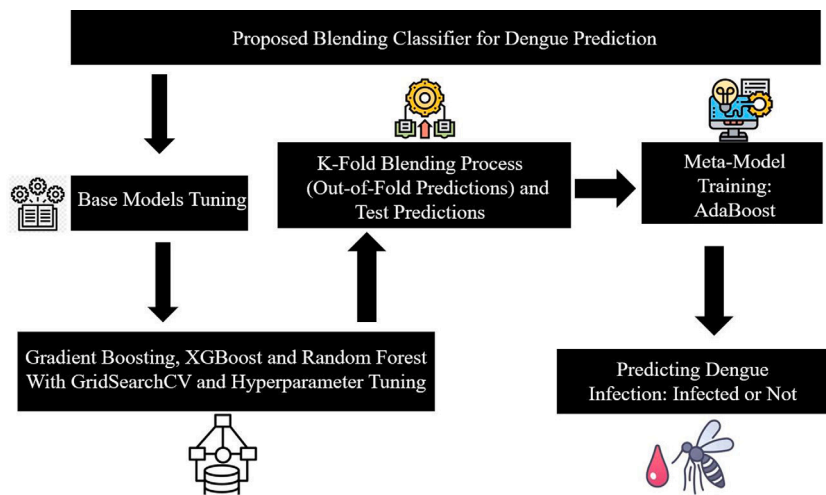


Fig. 9. Model Architecture of the Proposed Blending Classifier.

on the NS1 target: two in-context foundation models, TabPFN and TabICL [35]; three attention-based networks, FT-Transformer [36], SAINT [37], and TabNet [38]; and two further tabular deep-learning models, TabM [39] and TabR [40]. Each model was wrapped in a scikit-learn pipeline that fitted a “StandardScaler” ahead of the estimator, so standardization was learned from the training split alone and no test-set statistics leaked into fitting. Rather than cross-validation, all models were trained once on a fixed 90/10 train–test split, which is the same 117-patient test set used for the LLM comparison and evaluated on the held-out test set. No hyperparameter tuning was performed: TabPFN and TabICL are hyperparameter-free, while the remaining models (FT-Transformer, TabM, TabR, SAINT, and TabNet) ran at fixed default settings, with the gradient-trained networks using early stopping on an internal stratified validation split (15% of training data, patience 15).

3.9. Large language model training

The initial text-based design of Large Language Models (LLMs) has proven successful in analyzing tabular data. The architecture of LLMs can be adapted, or specialized fine-tuning techniques can be applied to enable these models to process structured rows and columns and discover complex patterns and relationships that traditional models typically miss. The ability of these models to achieve state-of-the-art performance on diverse tabular tasks such as classification, regression, and feature importance analysis, particularly on complex or high-dimensional datasets. Their success comes from their ability to learn intricate interactions within the data and leverage transferable

knowledge from vast pre-training corpora [41]. The performance of pre-trained LLMs can be improved for specific applications through fine-tuning on particular tabular datasets.

3.9.1. BERT

A pre-trained BERT tokenizer and sequence classification model (bert-base-cased) were retrieved from Hugging Face for fine-tuning on the employed Dengue dataset. The text data underwent preprocessing with a tokenization algorithm that implemented both truncation and padding, standardizing each sequence to a maximum of 256 characters. The hyperparameters included 20 training epochs, a batch size of 16, a learning rate of 1e-5, 100 warmup steps, a weight decay of 0.3, and early stopping with a patience of 5 epochs. The model was regularized using a hidden dropout and attention dropout probability of 0.2.

3.9.2. BioBERT

The Hugging Face BioBERT model (biobert-base-cased-v1.1) was used with its associated tokenizer for sequence classification tasks using Dengue patient data. The text column was tokenized in batches, with padding and truncation applied to a maximum sequence length of 256 tokens for consistent input formatting. The model hyperparameters included 20 training epochs, a batch size of 16, a learning rate of 1e-5, 100 warmup steps, a weight decay of 0.3, and early stopping with a patience of 5 epochs. The hidden and attention dropout probabilities were established at 0.2 to regularize the model and mitigate overfitting.

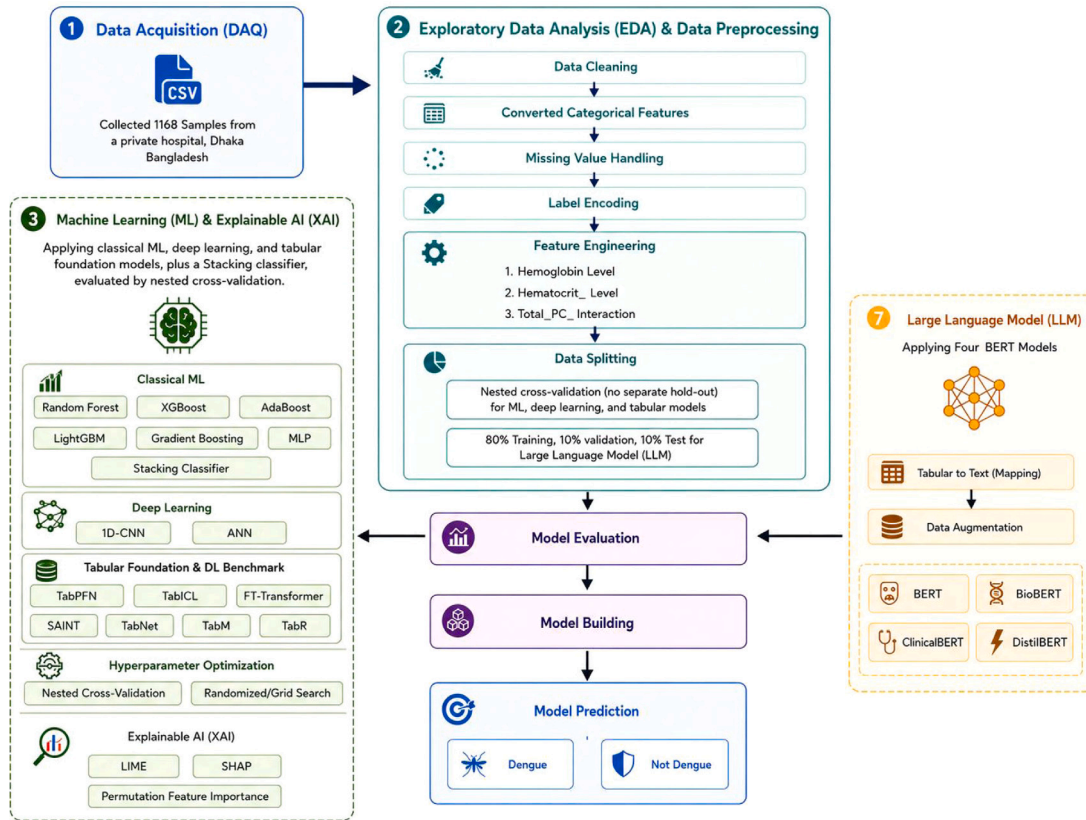


Fig. 10. Proposed Methodology Pipeline for Dengue Classification.

3.9.3. ClinicalBERT

The ClinicalBERT tokenizer and sequence classification model (clinical-bert-cased) were retrieved from Hugging Face for fine-tuning on our dataset. The text column underwent processing using a tokenization function that included padding and truncation to a maximum of 256 tokens, executed in a batched approach across the training, validation, and test datasets. The model was trained via the Hugging Face Trainer API, with parameters configured for 20 epochs, a batch size of 16, a learning rate of $1e-5$, 100 warmup steps, a weight decay of 0.3, and early stopping implemented with a patience of 5 epochs. Furthermore, the probabilities for hidden and attention dropout were established at 0.2 to regularize the model during training.

3.9.4. DistilBERT

We used a local version of the pre-trained 'distilbert-base-cased' model and its tokenizer from Hugging Face for the distilBERT model. The texts were processed using a specialized function that padded and shortened each sequence to a maximum of 256 tokens prior to batching and processing for training, validation, and test sets. The model training was conducted using Hugging Face's Trainer interface, consisting of 20 epochs, a batch size of 16, an initial learning rate of $1e-5$, a weight decay of 0.3, and 100 warmup steps, with early stopping activated after 5 epochs without improvement. A dropout probability of 0.2 was included in the model setup to enhance generalization.

3.10. Explainability analysis

Model behavior was characterized using three complementary attribution methods: LIME for local explanations, SHAP for global attributions, and PFI for ranking in the test set. All three were applied to the standardized feature space in which the models were trained. A fixed random seed (42) was used throughout for reproducibility.

3.11. Explainable AI

This subsection explains how explainable AI models were prepared in this work to interpret the automatic dengue prediction.

3.11.1. Local interpretable model-agnostic explanations (LIME)

Local Interpretations of Model-Agnostic Explanations (LIME) [42] is a local explainable artificial intelligence (XAI) method used to produce human-interpretable and instance-specific explanations for complex ML models. Local explanations were generated with LimeTabularExplainer, using the standardized training partition as the reference distribution, with continuous features discretized and all features reported for each explained instance. Because a single LIME explanation is local and sensitive to the sampled neighborhood, weights were averaged in absolute value over a random subsample of instances to give an approximate global view; both the individual explanations and the aggregate are reported.

3.11.2. SHAP

Shapley Additive Explanations (SHAP) [43] determines the exact impact of each feature on a particular prediction by applying game theory principles to model features as cooperative players who share the model output as their reward. SHAP values were computed to attribute each model's predicted probability of the positive class to the input features. For our model whose structure does not permit exact Shapley computation, including ensembles that combine heterogeneous base learners through a meta-learner, the model-agnostic KernelExplainer was used, which treats the model as a black box and estimates Shapley values by fitting a weighted local surrogate. To bound the cost of this sampling-based estimator, the background distribution was summarized by k -means clustering of the standardized training data ($k = 50$), 200 coalitions were sampled per instance, and values were computed for a random subsample of 150 observations.

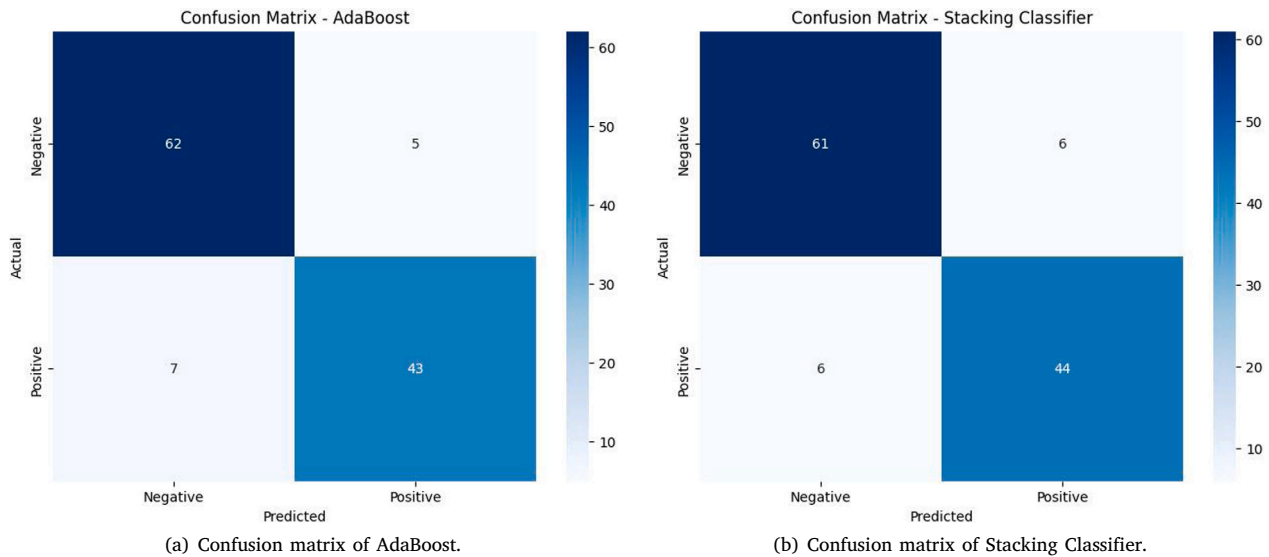


Fig. 11. Confusion matrix of AdaBoost and Stacking Classifier on the test set.

Table 1

Performance metrics of the applied ML models with hyperparameter tuning.

Model	Precision	Recall	F1 score	Accuracy (%)
LightGBM	0.796	0.860	0.826	84.62
Gradient Boosting	0.826	0.860	0.843	86.32
XGBoost	0.875	0.840	0.857	88.03
Blending Classifier	0.846	0.880	0.862	88.03
Random Forest	0.888	0.888	0.888	88.89
Bagging Classifier	0.849	0.900	0.873	88.89
AdaBoost	0.895	0.860	0.877	89.74
Stacking Classifier	0.880	0.880	0.880	89.74

Table 2

Performance metrics of the applied DL models with hyperparameter tuned.

Model	Precision	Recall	F1 score	Accuracy (%)
ANN	0.761	0.640	0.695	76.92
1D-CNN	0.755	0.680	0.715	76.92
MLP	0.795	0.780	0.787	82.05

3.11.3. Permutation feature importance

Permutation Feature Importance [44] is one of the global explainable AI (XAI) methods that dictate the importance of a model. Permutation feature importance was computed on the held-out test partition as the mean decrease in ROC-AUC when a single feature is randomly shuffled, over 50 permutations, with intervals given by the 2.5–97.5 percentiles of the resulting distribution. These intervals describe variability across shuffles and are not confidence intervals on a population quantity. Because permutation removes one feature at a time, the procedure understates the contribution of features that are collinear with, or deterministic functions of, other columns.

3.12. Workflow

Fig. 10 illustrates the comprehensive working sequences for the proposed dengue prediction system using private tabular health data. The process begins with data loading, and then we perform preprocessing, such as missing value handling and label encoding. Feature engineering includes medically guided transformations, such as categorizing hemoglobin and hematocrit levels and generating a total count-to-platelet count interaction feature. After splitting the data (90/10)

for machine learning models, deep learning, and tabular foundation models and (80/10/10) for large language models, various machine learning, deep learning, tabular foundation, and large language models are trained, evaluated, and interpreted using explainable AI techniques such as LIME, SHAP, and Permutation Feature Importance.

4. Results and discussion

4.1. Classification performance of machine learning, deep learning, and tabular foundation models

Fig. 11 shows the confusion matrices for the AdaBoost and Stacking classifiers on the test set. Both models classified the majority of cases correctly. AdaBoost correctly identified 62 of 67 negative cases and 43 of 50 positive cases, with 7 dengue-positive cases misclassified as negative (false negatives) and 5 false positives. The stacking classifier produced a slightly more balanced error profile, correctly identifying 61 negatives and 44 positives, with 6 false negatives and 6 false positives.

Fig. 12 shows the AdaBoost and stacking classifier's Receiver Operating Characteristic (ROC) curve. It shows a relation between the true positive rate (sensitivity) and the false positive rate. In an independent test cohort, the area under the curve (AUC) of AdaBoost is 0.93, showing a strong discriminative performance for the model. The stacking classifier showed better discriminative performance with an AUC of 0.94. A close-to-left-top corner of the curve indicates an ability to distinguish between the positive and negative classes. The diagonal is a random classifier that serves as a baseline (clearly, the AdaBoost model does much better than this baseline).

Performance metrics (precision, recall, F1 score, and accuracy) for the applied machine learning models on the proposed dengue classification task are shown in Table 1. With the highest F1 score (0.880) among the ML and DL models, the stacking classifier showed optimal results among all the ML and DL models. Almost all of the classifiers provide competitive results, with Random Forest scoring over 88%, as do bagging classifiers and blending classifiers. The stacking classifier was then evaluated under a nested cross-validation protocol with hyperparameters selected by inner cross-validation to showcase generalizability. 95% bootstrap confidence intervals (2000 resamples, seed 42) were applied. The model achieved a precision of 0.874 [0.861–0.887], a

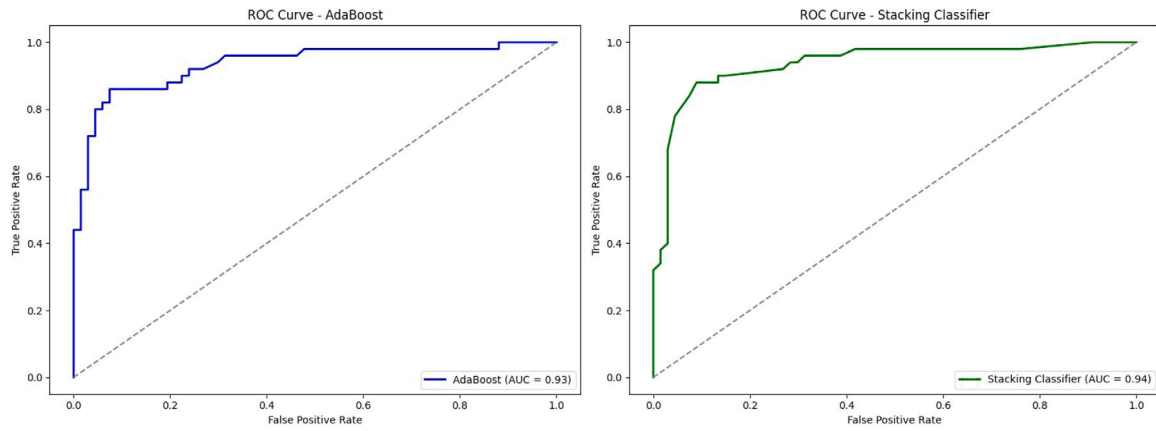


Fig. 12. ROC curve of AdaBoost and Stacking Classifier.

Table 3

Tabular foundation models on the identical held-out test set used for the LLMs ($n = 117$, $n_{pos} = 41$), with 95% bootstrap confidence intervals (2000 resamples, seed 42).

Model	Precision	Recall	F1 score	Accuracy (%)
TabPFN	0.8286 [0.6944, 0.9474]	0.7073 [0.5600, 0.8400]	0.7632 [0.6415, 0.8637]	84.62 [77.78, 90.60]
TabICL	0.8286 [0.6944, 0.9474]	0.7073 [0.5652, 0.8409]	0.7632 [0.6415, 0.8637]	84.62 [77.78, 90.60]
TabM	0.7692 [0.6279, 0.8974]	0.7317 [0.5946, 0.8600]	0.7500 [0.6349, 0.8451]	82.91 [76.07, 89.74]
FT-Transformer	0.8125 [0.6667, 0.9375]	0.6341 [0.4838, 0.7778]	0.7123 [0.5756, 0.8169]	82.05 [75.21, 88.89]
TabR	0.7879 [0.6452, 0.9200]	0.6341 [0.4872, 0.7778]	0.7027 [0.5789, 0.8095]	81.20 [74.36, 88.03]
SAINT	0.6275 [0.4912, 0.7593]	0.7805 [0.6512, 0.9000]	0.6957 [0.5757, 0.7963]	76.07 [68.38, 83.76]
TabNet	0.9091 [0.7000, 1.0000]	0.2439 [0.1163, 0.3810]	0.3846 [0.2041, 0.5424]	72.65 [64.96, 80.36]

recall of 0.873 [0.860–0.886], an F1 score of 0.873 [0.858–0.886], and an accuracy of 89.1 [87.7–90.4].

Table 2 showcases the performance metrics of the applied DL models with hyperparameter tuning (seed 42). Neural network-based models ANN and 1D-CNN, along with MLP, have relatively weaker performance (accuracy < 83%), which is still in line with the observation of ensemble methods and large language models being far more superior than standalone neural network models for this dataset.

Table 3 reports the tabular foundation models evaluated on the identical held-out test set used for the LLMs, enabling a direct, like-for-like comparison. On this common split, the strongest tabular models were TabPFN and TabICL (F1 0.7632, accuracy 84.62%). TabNet performed worst, with very low recall (0.2439). Notably, all tabular models fell well below the fine-tuned LLMs on the same test set: the best tabular F1 (0.7632) is substantially lower than that of BERT (Table 6). It indicates that the text-based representation captures signal beyond the structured CBC features for this task.

Table 4 represents the optimized hyperparameters and corresponding accuracies of the applied machine learning models with GridSearchCV. The AdaBoost model's maximum accuracy (89.74%) was reached with the SAMME.R algorithm, 0.1 learning rate, and 200 estimators. The next is the Random Forest model with 88.89% accuracy, with entropy for the criterion, max_depth = 8, and n_estimators = 100. Along with that, the stacking classifier with max_depth 8, max_features sqrt, and n_estimators 500. The Blending Classifier gave an accuracy of 88.03% with the algorithm selected as SAMME.R, a learning rate of 0.01, and n_estimators of 200. The Gradient Boosting and XGBoost models gave accuracies of 86.32% and 88.03%, respectively, using a max_depth of 3 with the colsample_bytree set to 1.0 for the XGBoost model. LightGBM (accuracy 84.62%) utilized a tree structure of greater size with 50 leaves, a learning rate of 0.01, and as many as 500 estimators. From these results, we can see that the best parameters differ significantly across boosting and ensemble models.

Table 5 summarizes the optimal hyperparameters and their accuracies corresponding to three deep learning models: MLP, ANN, and 1D-CNN. The best accuracy, 82.05%, is obtained on the MLP model

with parameters listed as ReLU activation, one hidden layer (size = 100), and optimizer: Adam. The ANN and 1D-CNN models, on the other hand, are close but slightly lower performing in terms of accuracy. Both employ ReLU activation and the Adam optimizer to achieve 76.92%, respectively. For the 1D-CNN, a few more architectural details are added with a kernel size of 3 and a pooling size of 2; we have only one hidden layer for the ANN.

4.2. Classification performance of large language models

Fig. 13 presents the confusion matrix and ROC curve for the BERT model on the held-out test set ($n = 117$). The model correctly classifies 73 of 76 negative and 39 of 41 positive cases, with 3 false positives and 2 false negatives. The ROC curve yields an AUC of 0.98, which indicates strong separability between the two classes.

Fig. 14 shows the training and validation loss (left) and validation metrics (right) across epochs. Training and validation loss decrease together without diverging, and the gap between them remains small throughout. It showcases that overfitting is controlled. Checkpoint selection based on validation F1 identified epoch 17 as the best epoch (dashed line); validation metrics stabilize well before this point. Validation recall is noticeably more variable than precision across epochs, consistent with the small number of positive cases in the validation set.

Fig. 15 shows the learning curve (left) and calibration curve (right) for BERT. In the learning curve, validation F1 rises steeply as training data is added and then improves more gradually, while the gap between training and validation F1 remains small and does not widen with increasing data. Horizontal-axis sizes refer to augmented training examples derived from the patient-level training split and do not represent independent patients. The calibration curve, computed with quantile binning, tracks the diagonal closely with a Brier score of 0.0394; the coarse binning reflects the limited size of the test set.

Table 6 reports precision, recall, F1 score, and accuracy for the four fine-tuned language models on the held-out test set, with 95%

Table 4
Best hyperparameters of the ML models.

Algorithm	Best hyperparameters	Accuracy (%)
LightGBM	'colsample_bytree': 1, 'learning_rate': 0.01, 'max_depth': 10, 'n_estimators': 500, 'num_leaves': 50, 'subsample': 0.8	84.62
Gradient Boosting	'learning_rate': 0.01, 'max_depth': 4, 'n_estimators': 500, 'subsample': 0.8	86.32
XGBoost	'colsample_bytree': 1.0, 'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 50, 'subsample': 0.8	88.03
Random Forest	'criterion': 'entropy', 'max_depth': 8, 'max_features': 'sqrt', 'n_estimators': 100	88.89
AdaBoost	'algorithm': 'SAMME.R', 'learning_rate': 0.1, 'n_estimators': 200	89.74
Blending Classifier	'algorithm': 'SAMME.R', 'learning_rate': 0.01, 'n_estimators': 200	88.03
Bagging Classifier	'estimator_max_depth': None, 'max_features': 0.8, 'max_samples': 0.5, 'n_estimators': 50	88.89
Stacking Classifier	'criterion': 'entropy', 'max_depth': 8, 'max_features': 'sqrt', 'n_estimators': 500	89.74

Table 5
Best hyperparameters of the DL models.

Algorithm	Best hyperparameters	Accuracy (%)
MLP	'activation': 'relu', 'alpha': 0.05, 'hidden_layer_sizes': (100), 'learning_rate': 'constant', 'solver': 'adam'	82.05
ANN	'activation': 'relu', 'hidden_layers': (128), 'learning_rate': 0.001, 'solver': 'adam', 'loss': 'binary_crossentropy'	76.92
1D-CNN	'activation': 'relu', 'kernel_size': 3, 'pool_size': (2), 'learning_rate': 0.001, 'solver': 'adam', 'loss': 'binary_crossentropy'	76.92

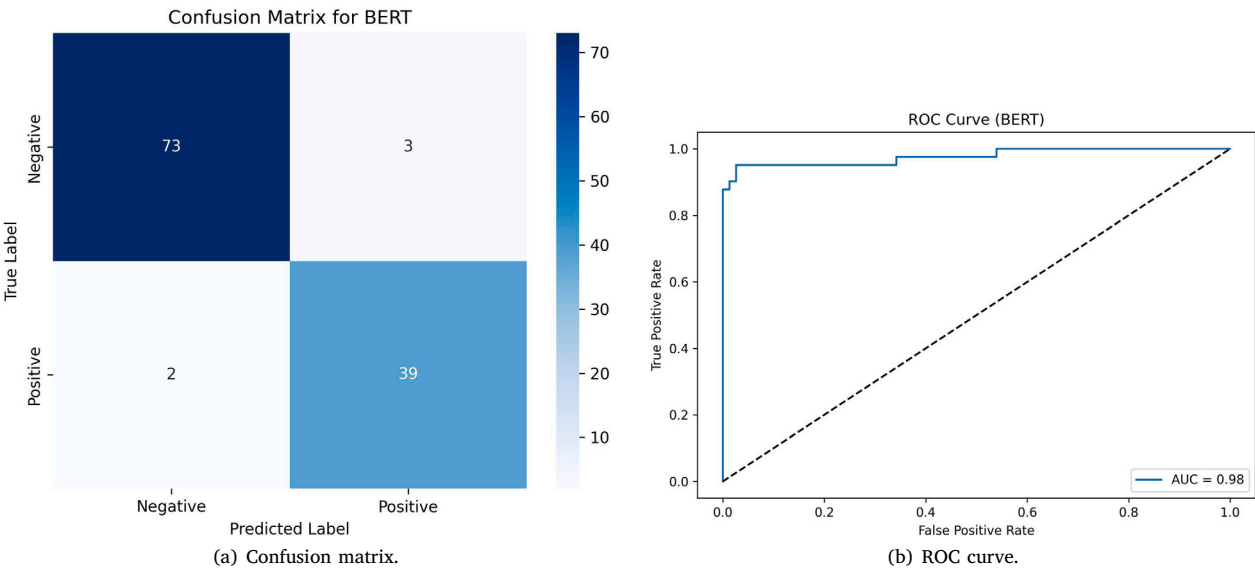


Fig. 13. Confusion matrix and ROC curve of the proposed BERT model.

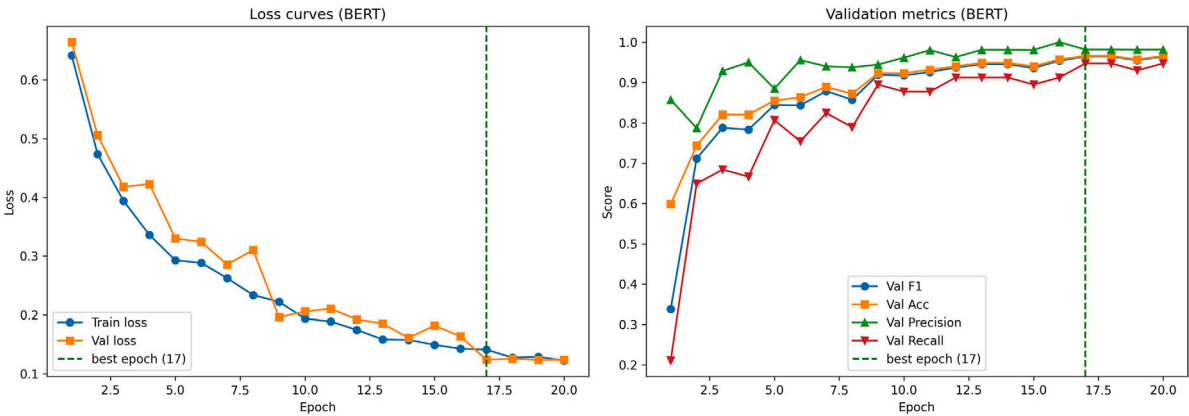


Fig. 14. Training and validation curves of the proposed BERT model.

bootstrap confidence intervals (2000 resamples) quantifying the uncertainty of each estimate. To prevent train–test leakage, data augmentation was applied to the training partition only, after the train/validation/test split. All four models achieve broadly comparable performance, with substantially overlapping confidence intervals across

every metric. BERT and BioBERT in particular are statistically indistinguishable. It indicates that biomedical-domain pretraining yields no detectable advantage over general-domain BERT for this specific classification task.

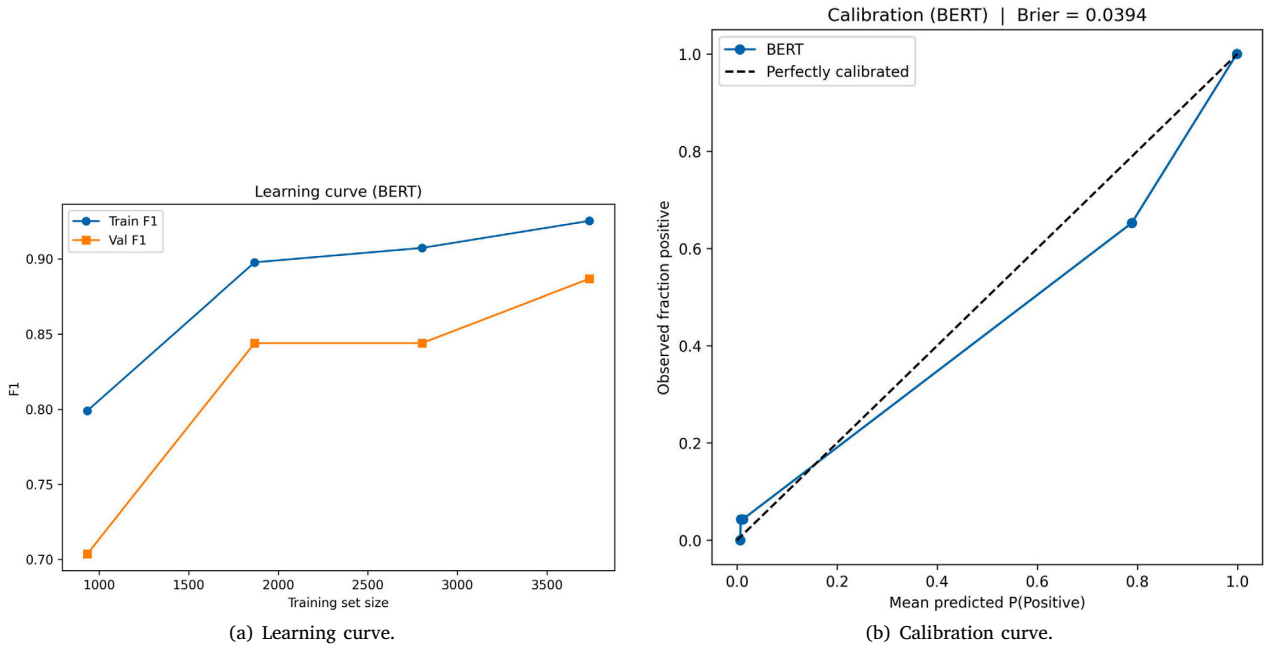


Fig. 15. Learning curve and calibration curve of the proposed BERT model.

Table 6

Performance metrics of the applied language models where values are test-set point estimates with 95% bootstrap confidence intervals (2000 resamples, seed 42).

Model	Precision [95% CI]	Recall [95% CI]	F1 score [95% CI]	Accuracy (%) [95% CI]
distilBERT	0.857 [0.744, 0.951]	0.878 [0.771, 0.974]	0.868 [0.783, 0.938]	90.60 [85.47, 95.73]
ClinicalBERT	0.947 [0.863, 1.000]	0.878 [0.771, 0.974]	0.911 [0.842, 0.972]	94.02 [89.74, 98.29]
BioBERT	0.973 [0.914, 1.000]	0.878 [0.771, 0.974]	0.923 [0.857, 0.977]	94.87 [90.60, 98.29]
BERT	0.929 [0.844, 1.000]	0.951 [0.878, 1.000]	0.940 [0.879, 0.987]	95.73 [91.45, 99.15]

Table 7

Additional clinical performance metrics on the held-out test set, with 95% bootstrap confidence intervals (2000 resamples, seed 42).

Model	Specificity [95% CI]	NPV [95% CI]	ROC-AUC [95% CI]	PR-AUC [95% CI]
distilBERT	0.921 [0.859, 0.975]	0.933 [0.873, 0.986]	0.973 [0.942, 0.995]	0.964 [0.923, 0.993]
ClinicalBERT	0.974 [0.933, 1.000]	0.937 [0.880, 0.987]	0.980 [0.957, 0.996]	0.972 [0.939, 0.994]
BioBERT	0.987 [0.959, 1.000]	0.938 [0.882, 0.987]	0.985 [0.963, 0.999]	0.979 [0.948, 0.997]
BERT	0.961 [0.913, 1.000]	0.973 [0.933, 1.000]	0.981 [0.951, 1.000]	0.977 [0.943, 0.999]
Ensemble LLM1	0.987 [0.958, 1.000]	0.938 [0.877, 0.987]	–	–
Ensemble LLM2	0.974 [0.932, 1.000]	0.937 [0.875, 0.987]	–	–

Table 8

Paired statistical comparison of BERT and BioBERT on the held-out test set ($n = 117$, $n_{pos} = 41$). Δ denotes the BERT – BioBERT difference. McNemar's test: exact binomial, $p = 1.0$ (5 discordant predictions).

Metric	BERT	BioBERT	Δ	95% CI on Δ	Test	p
AUC	0.981	0.985	−0.004	[−0.033, +0.019]	DeLong	0.708
F1	0.940	0.923	+0.017	[−0.043, +0.077]	Paired bootstrap	0.490
Accuracy	0.957	0.949	+0.009	[−0.034, +0.051]	Paired bootstrap	0.835
Precision	0.929	0.973	−0.044	[−0.121, +0.004]	Paired bootstrap	0.279
Recall	0.951	0.878	+0.073	[+0.000, +0.158]	Paired bootstrap	0.097

Table 7 reports specificity, negative predictive value (NPV), ROC-AUC, and PR-AUC, complementing the precision, recall, F1, and accuracy in Table 6; sensitivity and PPV are equivalent to the recall and precision reported there and are not repeated. All models achieved high specificity (0.921–0.987) and NPV (0.933–0.973). This showcases reliable identification of negative cases. For the four individual models, ROC-AUC was uniformly high (0.973–0.985); PR-AUC, the more informative summary under class imbalance ($n_{pos} = 41$ of 117, a positive prevalence of 0.35), ranged from 0.964 to 0.979, well above the no-skill baseline of 0.35. As the majority-voting ensembles output discrete labels rather than probabilities, ROC-AUC and PR-AUC are not defined

for them. Consistent with their lower recall, both ensembles had slightly lower NPV (0.937–0.938) than the strongest individual model (BERT, 0.973).

Table 8 presents a paired statistical comparison of BERT and BioBERT on the held-out test set, where Δ denotes the BERT – BioBERT difference. AUC was compared using DeLong's test, and F1, accuracy, precision, and recall were calculated using a paired bootstrap (2000 resamples) that resamples both models on the same test instances to preserve pairing. No metric reaches statistical significance ($p > 0.05$ throughout), and McNemar's test on hard predictions agrees (exact binomial, $p = 1.0$; 5 discordant pairs).

Table 9
Hyperparameters of the LLMs.

Algorithm	Best hyperparameters
BioBERT	'batch_size': 16, 'learning_rate': 1e-5, 'epochs': 20, 'patience': 5, 'weight_decay': 0.3, 'warmup_steps': 100
distilBERT	'batch_size': 16, 'learning_rate': 1e-5, 'epochs': 20, 'patience': 5, 'weight_decay': 0.3, 'warmup_steps': 100
BERT	'batch_size': 16, 'learning_rate': 1e-5, 'epochs': 20, 'patience': 5, 'weight_decay': 0.3, 'warmup_steps': 100
ClinicalBERT	'batch_size': 16, 'learning_rate': 1e-5, 'epochs': 20, 'patience': 5, 'weight_decay': 0.3, 'warmup_steps': 100

Table 10
Ensemble performance on the held-out test set, with 95% bootstrap confidence intervals (2000 resamples, seed 42).

Ensemble	Precision [95% CI]	Recall [95% CI]	F1 [95% CI]	Accuracy (%) [95% CI]
LLM1	0.973 [0.912, 1.000]	0.878 [0.767, 0.974]	0.923 [0.854, 0.975]	94.87 [90.61, 98.33]
LLM2	0.947 [0.865, 1.000]	0.878 [0.767, 0.974]	0.911 [0.836, 0.966]	94.02 [89.73, 97.40]

Table 11

Paired comparison of each ensemble against the strongest individual model (BERT, test F1 = 0.940) on the held-out test set. Δ denotes ensemble – BERT; negative values indicate the ensemble performed worse.

Ensemble	Metric	Δ (Ens – BERT)	95% CI on Δ	p
LLM1	F1	−0.017	[−0.077, +0.043]	0.490
LLM1	Accuracy	−0.009	[−0.051, +0.034]	0.835
LLM1	Precision	+0.044	[−0.004, +0.121]	0.279
LLM1	Recall	−0.073	[−0.158, +0.000]	0.097
LLM2	F1	−0.028	[−0.085, +0.021]	0.232
LLM2	Accuracy	−0.017	[−0.051, +0.017]	0.458
LLM2	Precision	+0.019	[−0.010, +0.078]	0.734
LLM2	Recall	−0.073	[−0.158, +0.000]	0.097

Table 9 presents the hyperparameter values together with their corresponding classification results for each large language model. The experiments ran with a 16 batch size, a 1e-5 learning rate, 20 training epochs, 5 early-stopping patience, a 0.3 weight-decay factor, dropout of 0.2, and 100 warm-up steps.

4.3. Ablation study

This subsection contains the ensemble LLM model's performance, the importance of feature engineering and text augmentation and external validation on our dataset.

4.3.1. Ensemble LLM classifier performance

Two ensemble classifiers were constructed by combining the predictions of three fine-tuned models each: ensemble LLM1 (BioBERT, BERT, and ClinicalBERT) and ensemble LLM2 (BioBERT, BERT, and distilBERT). For each test sample, the predicted class labels from the three constituent models were aggregated. We evaluated four voting strategies. Those are majority voting, weighted-average voting, median voting, and product-rule voting and selecting the strategy on the validation set and applying it once to the held-out test set. On the validation set, the four strategies produced identical hard predictions, so majority voting was adopted for both ensembles. Since the ensembles are constructed using the trained LLMs, no additional training was conducted. So, we did not report any training and learning curves.

Fig. 16 shows the confusion matrices for the two ensemble classifiers on the held-out test set ($n = 117$). Ensemble LLM1 correctly classified 75 of 76 negative and 36 of 41 positive cases (1 false positive, 5 false negatives), and ensemble LLM2 correctly classified 74 of 76 negative and 36 of 41 positive cases (2 false positives, 5 false negatives).

Table 10 reports the performance of the two ensemble classifiers on the held-out test set. Both ensembles selected majority voting on the validation set; the four voting rules examined (majority, weighted-average, median, and product) produced identical hard predictions on the validation set, so the choice among them did not affect the decision. Because majority voting yields discrete class labels rather than probabilities, AUC is not defined for the ensembles. Clinical metrics

for the ensembles are included in Table 7. Ensemble LLM1 and LLM2 reproduce the exact label sets of BioBERT and ClinicalBERT, respectively, on this test set; the matching values are expected, not duplicated entries.

Table 11 compares each ensemble against the strongest individual model on the test set (BERT). Neither ensemble surpasses BERT: the F1 differences were negative (−0.017 for LLM1 and −0.028 for LLM2) and no metric reached statistical significance under either the paired bootstrap or McNemar's test ($p > 0.05$ throughout; all differences remained non-significant after Holm correction, $p_{\text{Holm}} > 0.9$).

4.3.2. Impact of feature engineering

Table 12 reports the contribution of the engineered features (Hemoglobin_Level, Hematocrit_Level, and Total_PC_Interaction). Both models were retrained and re-tuned under identical protocols on the raw feature set alone and on the full engineered set. Feature engineering improves F1 for both models, by 0.029 for XGBoost and by 0.071 for the stacking classifier. The gain is larger for the stacking classifier and arises almost entirely through recall: without the engineered features the model is conservative. It achieved high precision (0.923) at the cost of missing positive cases, whereas with them, recall rises from 0.720 to 0.880 while precision falls from 0.923 to 0.880. XGBoost shows the same directional pattern but a smaller shift, gaining both precision and recall modestly.

4.3.3. Impact of text data augmentation

Table 13 presents the impact of text augmentation. Each model was trained once on the original training split and later with augmentation applied to the training partition and evaluated on the same held-out test set ($n = 117$). Augmentation improved F1 for every model, with corresponding gains in accuracy and, in most cases, both precision and recall. The improvement was largest for BERT, whose F1 increased from 0.660 to 0.940.

4.3.4. External validation

To assess generalization beyond our primary dataset, we re-ran the external dataset from Riya et al. [14] using the same model configurations and hyperparameters employed in our study. Table 14 reports accuracy and macro-averaged F1 for Random Forest and XGBoost alongside the values reported in the original work. Random Forest reproduced the reported accuracy exactly (90.62%) with a marginally higher macro F1 (0.9015 vs. 0.8953), while XGBoost exceeded the reported figures on both metrics (96.88% vs. 92.71% accuracy; 0.9646 vs. 0.9198 macro F1).

4.4. Explainable AI analysis

Fig. 17 shows a LIME explanation for a representative test instance. For this patient, an above-average hemoglobin value and the corresponding elevated hemoglobin-level category contributed most strongly toward the NS1-positive prediction, followed by a below-average total-count/platelet interaction term; a below-average total count was the

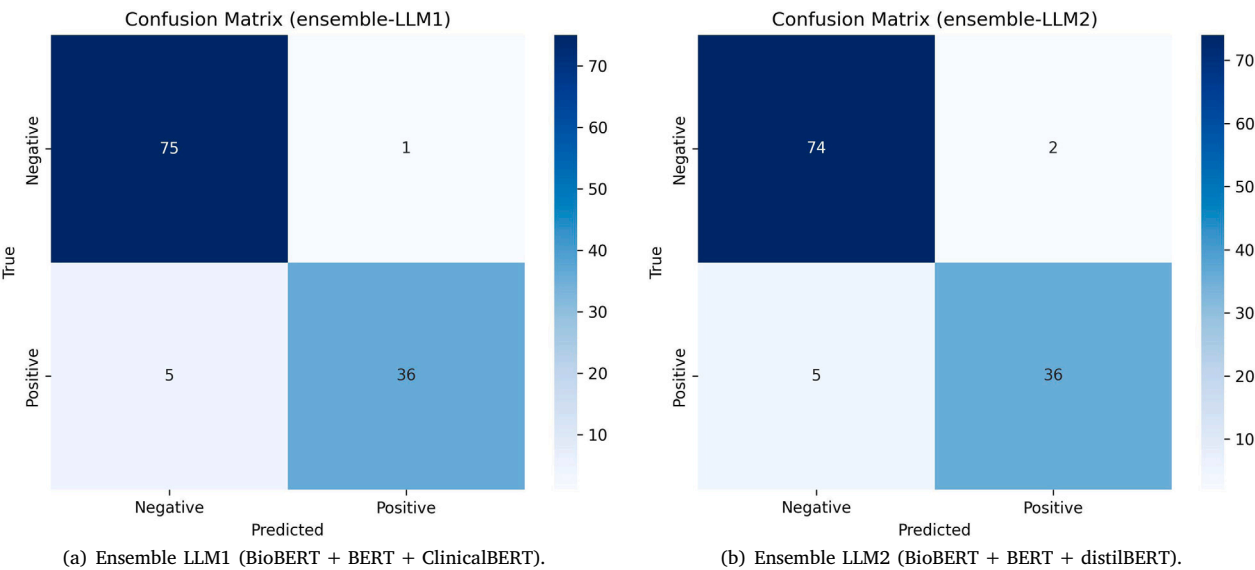


Fig. 16. Confusion matrices of the two ensemble classifiers.

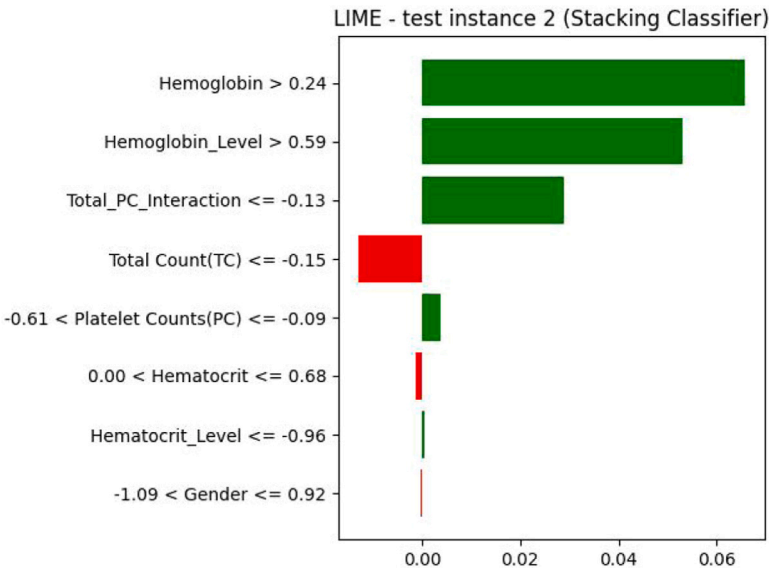


Fig. 17. LIME explanation for a representative test instance under the stacking classifier.

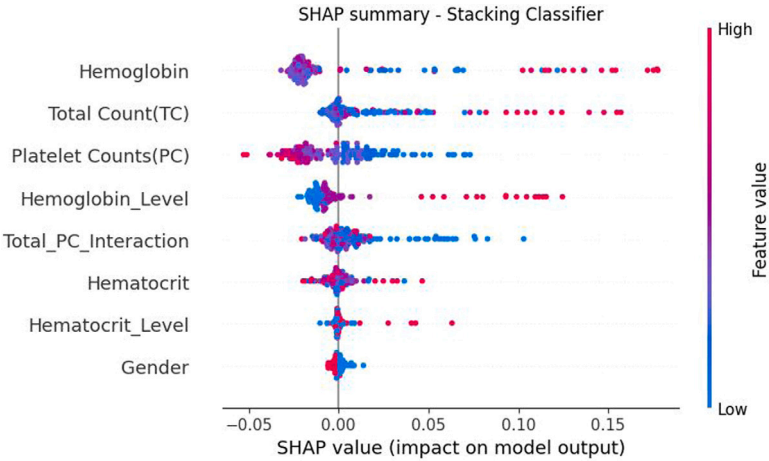


Fig. 18. Global SHAP summary for the stacking classifier.

Table 12

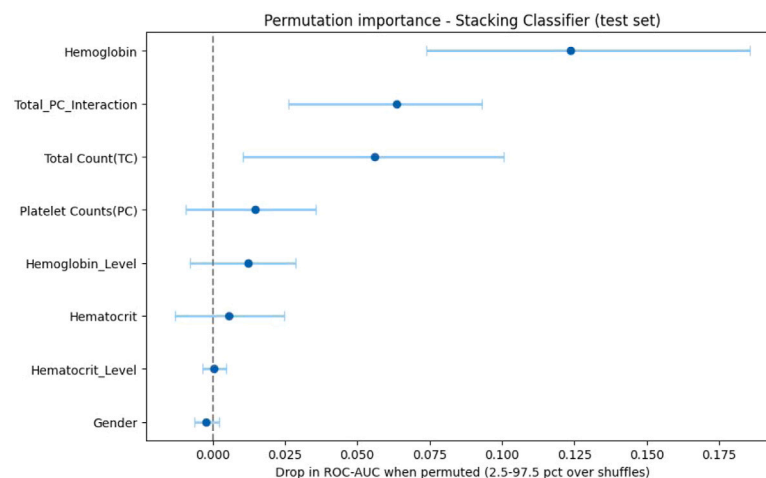
Impact of feature engineering (FE) on model performance. $\Delta F1$ is the improvement in F1 from feature engineering. All metrics are computed on the same held-out test partition ($n = 117$).

Model	Setting	Precision	Recall	F1	Accuracy (%)	$\Delta F1$
XGBOOST	Without FE	0.837	0.820	0.828	85.47	+0.029
	With FE	0.875	0.840	0.857	88.03	
Stacking Classifier	Without FE	0.923	0.720	0.809	85.47	+0.071
	With FE	0.880	0.880	0.880	89.74	

Table 13

Impact of text augmentation on model performance. Values are test-set point estimates with 95% bootstrap confidence intervals (2000 resamples, seed 42). $\Delta F1$ is the improvement in F1 from augmentation.

Model	Setting	Precision [95% CI]	Recall [95% CI]	F1 [95% CI]	Accuracy (%) [95% CI]	$\Delta F1$
BERT	Without aug.	0.571 [0.438, 0.698]	0.781 [0.647, 0.903]	0.660 [0.538, 0.763]	71.79 [63.33, 80.34]	+0.280
	With aug.	0.929 [0.844, 1.000]	0.951 [0.878, 1.000]	0.940 [0.879, 0.987]	95.73 [91.45, 99.15]	
BioBERT	Without aug.	0.667 [0.514, 0.804]	0.683 [0.531, 0.822]	0.675 [0.540, 0.783]	76.92 [69.26, 84.64]	+0.248
	With aug.	0.973 [0.914, 1.000]	0.878 [0.771, 0.974]	0.923 [0.857, 0.977]	94.87 [90.60, 98.29]	

**Fig. 19.** Permutation feature importance for the stacking classifier, computed on the test partition.**Table 14**

External validation of Random Forest and XGBoost. Accuracy and macro-averaged F1 obtained by running the source study's dataset under our model configuration, compared against the values reported in the original work.

Model	Accuracy (%)		Macro F1	
	Ours	Reported	Ours	Reported
Random Forest	90.62	90.62	0.9015	0.8953
XGBoost	96.88	92.71	0.9646	0.9198

only feature pushing appreciably in the opposite direction, and hematocrit, hematocrit level, and gender contributed negligibly. LIME thus attributes the individual prediction primarily to the same hematological features that dominate the global analysis, although the direction of the leading contribution is patient-specific and should not be read as a global effect.

Fig. 18 presents the global SHAP summary for the stacking classifier. Hemoglobin, total count, and platelet count show the widest SHAP value distributions, identifying them as the most influential features on model output, followed by the engineered hemoglobin-level and total-count/platelet interaction features. The coloring indicates the direction of each effect: high hemoglobin and high total count values are associated with positive SHAP contributions, whereas high platelet counts shift the model output downward, consistent with the thrombocytopenia characteristic of dengue infection.

Fig. 19 showcases permutation feature importance for the stacking classifier. Hemoglobin is by far the most important feature (mean drop

in ROC-AUC 0.124), followed by the total-count/platelet interaction (0.064) and total count (0.056); these three are the only features whose intervals exclude zero. Platelet count, hemoglobin level, hematocrit, hematocrit level, and gender all have intervals overlapping zero. It displays that shuffling them individually does not measurably degrade discrimination.

While hemoglobin and total count rank among the most important features under all three methods, the rankings are not identical in detail. Platelet count is the third-widest contributor under SHAP yet has a permutation-importance interval overlapping zero, and the engineered hemoglobin-level feature shows the same pattern. This divergence has a concrete cause rather than being an artifact of method choice: Total_PC_Interaction is the product of total count and platelet count, and Hemoglobin_Level is a deterministic recoding of hemoglobin and gender. Permuting either feature in isolation leaves its information available to the model through the correlated column, and the measured drop in ROC-AUC understates its contribution. SHAP, which distributes credit across features for each prediction rather than removing them one at a time, is less affected by this. The two methods measure different quantities in principle. SHAP attributes the model's output, and permutation importance the loss in discriminative performance so exact rank concordance is not expected. We treat the agreement on the leading hematological predictors as the robust finding and interpret the derived and interaction features as redundant encodings of that same signal rather than as independent contributors.

A comparative analysis of the proposed dengue prediction system with related studies is summarized in Table 15 based on sample size,

Table 15
Comparison of the proposed dengue classification system with related studies.

Study	Data Source and Sample Size	Best Model	Accuracy (%)	F1 Score
[15]	Local Hospitals of Dhaka and Chittagong, 209	Decision Tree	79%	0.790
[7]	UCI, 300	Fuzzy K-Nearest Neighbor	85.01%	N/A
[8]	Health Department in Malaysia, 84	SVM	85%	0.760
[18]	Five Cities of Malaysia, 1300	SVM	70%	0.600
[9]	Three local hospitals of Recife, 102	ANN	86%	N/A
[21]	Honduran, 548	Logistic Regression	69.2%	N/A
[14]	Dhaka, 322	Stacking Classifier	96.88%	0.965 (macro f1)
Proposed	Local Hospital of Dhaka, 1168	Stacking Classifier	89.74%	0.880
Proposed	Local Hospital of Dhaka, 1168	BERT	95.73%	0.940

data source, and performance metrics. The proposed BERT model, trained on the private dataset from a Dhaka hospital, achieved the highest accuracy (95.73%) and F1-score (0.940). This is one of the few studies to apply a large language model (LLM) and ensemble of various LLM techniques for classification employing a tabular dataset. This circumstance demonstrates the effectiveness of the proposed method in both predictive performance and generalizability, especially with a relatively large dataset of 1168 samples.

Several inherent limitations to our work need to be acknowledged, the first being that the private dataset, while comprehensive, is smaller than many publicly available dengue classification datasets. Although we attempted exhaustive feature engineering, not all the latent factors that might influence dengue diagnosis may have been captured fully. Implementing nested cross-validation and repeated stratified k-fold cross-validation was not computationally feasible due to the substantial training time and computational resources required for repeatedly fine-tuning transformer-based models. Our study is a retrospective clinical data review, and further prospective validation is needed to confirm the effectiveness and robustness of the models we propose here. These limitations confirm areas for improvement and further research directions.

5. Conclusion

Numerous researchers have dedicated their efforts to developing techniques for detecting dengue in recent years, and it is unexplored which classifier is effective for detecting dengue that effectively utilizes a minimal set of features and underlying driving factors that impact the model for its decision-making process, as it is important for a country like Bangladesh, where healthcare data is frequently inconsistent or inadequately maintained. This work proposed a private dataset, including 1168 samples of suspected patients with dengue whose key features were obtained from CBC data. The employed dataset suffers from a variety of missing features, so more advanced feature engineering was required to improve the predictive power of our data. Different machine learning, deep learning, tabular benchmark and large language models were evaluated, with hyperparameters tuned by GridSearchCV and Keras Tune for the models. Clinically relevant, gender-specific categorical features for hemoglobin and hematocrit levels and a total count-to-platelet count interaction feature were introduced, improving the precision and reliability of dengue diagnosis. The stacking classifier provided the optimal results among all ML and DL models, as it effectively differentiates between positive and negative dengue cases. Structured dengue patient data were converted into natural-language text sequences using pandas, enabling transformer LLM models, e.g., BERT and BioBERT, to perform classification on the hematological profile. The BERT showcased the best performance among all the models, although it is substantially more computationally expensive compared to ensemble models. XAI methods were incorporated into the workflow to give interpretability and clarify the predictions made by models. This process helps provide instance-specific explanations through local and global methods, including LIME, SHAP, and PFI, to understand the decision-making process of a complex model. This study is limited to employing dengue data from only the capital city of Bangladesh, Dhaka.

In the future, this dataset can be enhanced with additional samples from a larger group of patients from different regions of Bangladesh to develop more accurate and generalizable models. The plans involve creating a web application that scans patient medical reports and provides customized medication recommendations based on predictive outcomes. Federated learning frameworks provide a potential path forward for applications where it is necessary to maximize the use of sensitive patient information without putting individuals at risk by sharing their raw data; this can be done with model updates that keep relevant clinical accuracy while ensuring confidentiality. The future article can employ multimodal learning datasets, CBC and viral RNA-based tests, and optimized learning models with data pruning and quantization for dengue classification. This work will facilitate scalable, interpretable, and privacy-preserving systems of dengue detection.

CRediT authorship contribution statement

Muhammad Zawad Mahmud: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md. Shihab Reza:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation. **Samiha Islam:** Writing – original draft, Validation, Software, Methodology, Investigation, Data curation. **Shahran Rahman Alve:** Writing – original draft, Visualization, Validation, Methodology. **Riasat Khan:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Funding

The authors did not receive any funding for this investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code and dataset of the investigation is available at [Dengue_ML_Research](#).

References

[1] Khan MB, Yang Z-S, Lin C-Y, Hsu M-C, Urbina AN, Assavalapsakul W, Wang W-H, Chen Y-H, Wang S-F. Dengue overview: An updated systemic review. *J Infect Public Health* 2023;16:1625–42. <http://dx.doi.org/10.1016/j.jiph.2023.08.001>, URL <https://www.sciencedirect.com/science/article/pii/S1876034123002587>.

[2] Bäck AT, Åke Lundkvist. Dengue viruses – an overview. *Infect Ecol & Epidemiology* 2013;3.

[3] European Centre for Disease Prevention and Control. Dengue Monthly Update. 2024, URL <https://www.ecdc.europa.eu/en/dengue-monthly>, (Accessed 11 November 2024).

[4] Bonna AS, Pavel SR, Mehjabin T, Ali M. Dengue in Bangladesh. *IJID One Health* 2023;1. <http://dx.doi.org/10.1016/j.ijidoh.2023.100001>, URL <https://www.sciencedirect.com/science/article/pii/S294991512300001X>.

- [5] Salje H, Paul KK, Paul R, Rodriguez-Barraquer I, Rahman Z, Alam MS, Rahman M, Al-Amin HM, Heffelfinger J, Gurley E. Nationally-representative serostudy of dengue in Bangladesh allows generalizable disease burden estimates. *Elife* 2019;8.
- [6] Ali N. The recent burden of dengue infection in Bangladesh: A serious public health issue. *J Infect Public Health* 2024;17:226–8.
- [7] Jannah A, Husaini A, Ichsan A, Azhari M. Implementation of Fuzzy K-Nearest Neighbor Method in Dengue Disease Classification. *Hanif J Inf Syst* 2024;1:49–54.
- [8] Nordin N, Sobri NM, Ismail N, Zulkifli S, Abd Razak N, Mahmud M. The classification performance using support vector machine for endemic dengue cases. In: *Journal of physics: conference series*. vol. 1496, IOP Publishing; 2020.
- [9] Davi C, Pastor A, Oliveira T, de Lima Neto FB, Braga-Neto U, Bigham AW, Bamshad M, Marques ET, Aciole-Santos B. Severe dengue prognosis using human genome data and machine learning. *IEEE Trans Biomed Eng* 2019;66:2861–8.
- [10] Dang AT, Dang TN. Climate-Driven Dynamics of Infectious Diseases: Machine Learning-Based Predictive Modeling in Ba Ria-Vung Tau, Vietnam. *SSRN Electron J* 2025.
- [11] Lu X, Teh SY, Tay CJ, Abu Kassim NF, Fam PS, Soewono E. Application of multiple linear regression model and long short-term memory with compartmental model to forecast dengue cases in Selangor, Malaysia based on climate variables. *Infect Dis Model* 2024;10:240–56.
- [12] Gupta M, Bhattachakosol P. A Bayesian regularization intelligent computing scheme for the fractional dengue virus model. *Egypt Inform. J* 2025;29.
- [13] Dey SK, Rahman MM, Howlader A, Siddiqi UR, Uddin KMM, Borhan R, Rahman EU. Prediction of dengue incidents using hospitalized patients, metrological and socio-economic data in Bangladesh: A machine learning approach. *PLoS One* 2022;17.
- [14] Riya NJ, Chakraborty M, Khan R. Artificial intelligence based early detection of dengue using CBC data. *IEEE Access* 2024.
- [15] Sarma D, Hossain S, Mittra T, Bhuiya MAM, Saha I, Chakma R. Dengue prediction using machine learning algorithms. In: *IEEE 8th r10 humanitarian technology conference*. IEEE; 2020, p. 1–6.
- [16] Akter T, Islam MT, Hossain MF, Ullah MS. A Comparative Study between Time Series and Machine Learning Technique to Predict Dengue Fever in Dhaka City. *Discrete Dyn Nat Soc* 2024;2024.
- [17] Majeed MA, Shafri HZM, Zulkafli Z, Wayayok A. A deep learning approach for dengue fever prediction in Malaysia using LSTM with spatial attention. *Int J Environ Res Public Health* 2023;20.
- [18] Salim NAM, Wah YB, Reeves C, Smith M, Yaacob WFW, Mudin RN, Dapari R, Sapri NNFF, Haque U. Prediction of dengue outbreak in Selangor Malaysia using machine learning techniques. *Sci Rep* 2021;11.
- [19] Chaw JK, Chaw SH, Quah CH, Sahrani S, Ang MC, Zhao Y, Ting TT. A predictive analytics model using machine learning algorithms to estimate the risk of shock development among dengue patients. *Heal Anal* 2024;5.
- [20] Mello-Román JD, Mello-Román JC, Gómez-Guerrero S, García-Torres M. Predictive models for the medical diagnosis of dengue: A case study in Paraguay. *Comput Math Methods Med* 2019;2019. <http://dx.doi.org/10.1155/2019/7307803>, arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1155/2019/7307803>, URL <https://onlinelibrary.wiley.com/doi/abs/10.1155/2019/7307803>.
- [21] Fernández E, Smieja M, Walter SD, Loeb M. A predictive model to differentiate dengue from other febrile illness. *BMC Infect Dis* 2016;16.
- [22] Prome SS, Basak T, Plabon TI, Khan R. Prediction of Dengue Cases in Bangladesh using Explainable Machine Learning Approach. In: *International conference on inventive computation technologies*. IEEE; 2024.
- [23] Schechter AN. Hemoglobin research and the origins of molecular medicine. *Blood* 2008;112:3927–38, URL <https://api.semanticscholar.org/CorpusID:1728121>.
- [24] Billett HH. Hemoglobin and Hematocrit. In: Walker HK, Hall WD, Hurst JW, editors. *Clinical methods: the history, physical, and laboratory examinations*. third ed.. Boston: Butterworths; 1990, URL <https://www.ncbi.nlm.nih.gov/books/NBK259/>.
- [25] Kishimoto S, Maruhashi T, Kajikawa M, Matsui S, Hashimoto H, Takaeko Y, Harada T, Yamaji T, Han Y, Kihara Y, Chayama K, Goto C, Yusoff F, Nakashima A, Higashi Y. Hematocrit, hemoglobin and red blood cells are associated with vascular function and vascular structure in men. *Sci Rep* 2020;10. <http://dx.doi.org/10.1038/s41598-020-68319-1>.
- [26] Yang N-C, Ismail H. Robust intelligent learning algorithm using random forest and modified-independent component analysis for PV fault detection: In case of imbalanced data. *IEEE Access* 2022;10:41119–30. <http://dx.doi.org/10.1109/ACCESS.2022.3166477>.
- [27] Chen T, Guestrin C. XGBoost: A scalable tree boosting system. In: *International conference on knowledge discovery and data mining*. 2016, p. 785–94.
- [28] Friedman JH. Greedy function approximation: A gradient boosting machine. *Ann Statist* 2001;29:1189–232.
- [29] Yulianto A, Sukarno P, Suwastika NA. Improving adaboost-based intrusion detection system (IDS) performance on CIC IDS 2017 dataset. In: *Journal of physics: conference series*. vol. 1192, IOP Publishing; 2019.
- [30] Zhang J, Mucs D, Norinder U, Svensson F. LightGBM: An effective and scalable algorithm for prediction of chemical toxicity—application to the Tox21 and mutagenicity data sets. *J Chem Inf Model* 2019;59:4150–8.
- [31] Al Bataineh A, Kaur D, Jalali SMJ. Multi-layer perceptron training optimization using nature inspired computing. *IEEE Access* 2022;10:36963–77. <http://dx.doi.org/10.1109/ACCESS.2022.3164669>.
- [32] Kiranyaz S, Avci O, Abdeljaber O, Ince T, Gabbouj M, Inman DJ. 1D convolutional neural networks and applications: A survey. *Mech Syst Signal Process* 2021;151. <http://dx.doi.org/10.1016/j.ymssp.2020.107398>, URL <https://www.sciencedirect.com/science/article/pii/S0888327020307846>.
- [33] LeCun Y, Bengio Y, Hinton G. Deep learning. *Nature* 2015;521:436–44.
- [34] Hollmann N, Müller S, Eggensperger K, Hutter F. TabPFN: A transformer that solves small tabular classification problems in a second. In: *Annual conference on neural information processing systems*. 2022.
- [35] Qu J, Holzmann D, Varoquaux G, Morvan ML. Tablic: A tabular foundation model for in-context learning on large data. 2025, arXiv preprint [arXiv:2502.05564](https://arxiv.org/abs/2502.05564).
- [36] Gorishniy Y, Rubachev I, Khrulkov V, Babenko A. Revisiting deep learning models for tabular data. *Adv Neural Inf Process Syst* 2021;34:18932–43.
- [37] Somepalli G, Goldblum M, Schwarzschild A, Bruss CB, Goldstein T. Saint: Improved neural networks for tabular data via row attention and contrastive pre-training. 2021, arXiv preprint [arXiv:2106.01342](https://arxiv.org/abs/2106.01342).
- [38] Arik SÖ, Pfister T. Tabnet: Attentive interpretable tabular learning. In: *AAAI conference on artificial intelligence*. vol. 35, 2021, p. 6679–87.
- [39] Gorishniy Y, Kotelnikov A, Babenko A. Tabm: Advancing tabular deep learning with parameter-efficient ensembling. In: *International conference on learning representations*. vol. 2025, 2025, p. 77899–935.
- [40] Gorishniy Y, Rubachev I, Kartashev N, Shlenskii D, Kotelnikov A, Babenko A. Tabr: Tabular deep learning meets nearest neighbors. In: *International conference on learning representations*. vol. 2024, 2024, p. 18209–49.
- [41] Azad MS, Leon SI, Khan R, Mohammed N, Momen S. SAD: Self-assessment of depression for Bangladeshi university students using machine learning and NLP. *Array* 2024. <http://dx.doi.org/10.1016/j.array.2024.100372>.
- [42] Ribeiro MT, Singh S, Guestrin C. “Why should I trust you?” Explaining the predictions of any classifier. In: *International conference on knowledge discovery and data mining*. 2016, p. 1135–44.
- [43] Lundberg SM, Lee S-I. A Unified Approach to Interpreting Model Predictions. In: *Advances in neural information processing systems*. vol. 30, 2017, p. 4765–74.
- [44] Fumagalli F, Muschalik M, Hüllermeier E, Hammer B. Incremental permutation feature importance (IPFI): towards online explanations on data streams. *Mach Learn* 2023;112:4863–903.